

AdaPreLoRA: Adafactor Preconditioned Low-Rank Adaptation

Ziyun Liu Fengmiao Bian* Jian-Feng Cai*

Department of Mathematics
The Hong Kong University of Science and Technology
zliueq@connect.ust.hk {mafmbian, jfcai}@ust.hk

Abstract

Low-Rank Adaptation (LoRA) reparameterizes a weight update as a product of two low-rank factors, but the Jacobian J_G of the generator mapping the factors to the weight matrix is rank-deficient, so the factor-space preconditioner $J_G^* \mathcal{F}_t J_G$ induced by any $\mathbf{W}$ -space preconditioner $\mathcal{F}_t$ is singular, and consequently the standard chain rule cannot be uniquely inverted to map a preconditioned $\mathbf{W}$ -space direction back to a factor-space update. We cast existing LoRA optimizers in a unified framework parameterized by two choices: (i) which invertible surrogate for $J_G^* \mathcal{F}_t J_G$ to use, and (ii) which $\mathcal{F}_t$ on $\mathbf{W}$ to use. Existing methods occupy four families along these axes: factor-space adaptive updates, block-diagonal surrogates for $J_G^* J_G$, Frobenius-residual pseudoinverse methods, and Riemannian manifold constraint. Within this design space, a gradient-statistics-aware $\mathcal{F}_t$ paired with a closed-form factor-space solve at $\mathcal{O}((m+n)r)$ memory remains underexplored. We propose **AdaPreLoRA**, which fills this gap by adopting the Adafactor diagonal Kronecker preconditioner $\mathcal{H}_t$ on $\mathbf{W}$ and selecting from the resulting factor-space solution family the element minimizing an $\mathcal{H}_t$ -weighted imbalance between the two factor contributions; by construction, the resulting factor update is the closest LoRA approximation to the preconditioned $\mathbf{W}$ -space direction under the $\mathcal{H}_t$ -weighted norm. Across GPT-2 (E2E), Mistral-7B and Qwen2-7B (GLUE, ARC, GSM8K), and diffusion-model personalization, AdaPreLoRA is competitive with or improves over a representative set of LoRA optimizers while keeping peak GPU memory at the LoRA optimizer level.

1 Introduction

Fine-tuning large pretrained models [2, 18, 35] for downstream tasks is increasingly bottlenecked by the cost of full-parameter updates, motivating parameter-efficient fine-tuning (PEFT). Low-Rank Adaptation (LoRA) [14] has become the standard PEFT method: it freezes the pretrained weight $\mathbf{W}_0$ and reparameterizes its update as a product $\mathbf{B}\mathbf{A}$ with $\mathbf{B} \in \mathbb{R}^{m \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times n}$, $r \ll \min(m, n)$, reducing trainable parameters and optimizer state from $\mathcal{O}(mn)$ to $\mathcal{O}((m+n)r)$. A growing line of work [11, 21, 32, 33, 37–40] extends this template with refined optimizers in pursuit of full-fine-tuning quality at LoRA’s memory budget.

Despite this progress, optimizing in the factor space $[\mathbf{B}, \mathbf{A}]$ rather than directly in $\mathbf{W}$ raises a fundamental obstruction (§ 2): writing $\mathcal{G} : [\mathbf{B}, \mathbf{A}] \mapsto \mathbf{B}\mathbf{A}$ for the map generating the factors to the weight matrix, its Jacobian J_G is rank-deficient because $\mathcal{G}$ has a built-in redundancy under the gauge reparameterization $(\mathbf{B}, \mathbf{A}) \mapsto (\mathbf{B}\mathbf{C}, \mathbf{C}^{-1}\mathbf{A})$ for any invertible $\mathbf{C}$. Since practical gradient-statistical preconditioners are typically approximations to the Fisher information in the parameter space being

*Corresponding authors.

optimized, the relevant preconditioner in factor space is the Fisher information with respect to $[\mathbf{B}, \mathbf{A}]$. By the chain rule, this operator must take the form $J_G^* \mathcal{F}_t J_G$, where $\mathcal{F}_t$ is the corresponding gradient-statistical preconditioner in $\mathbf{W}$ -space. Because $J_G^* \mathcal{F}_t J_G$ is singular, it cannot be uniquely inverted to map a $\mathbf{W}$ -space preconditioned direction back to a factor-space update. Existing LoRA optimizers respond to this obstruction along several directions. Cheap factor-space schemes preserve the $\mathcal{O}((m+n)r)$ memory budget but discard the gradient-statistics structure on $\mathbf{W}$, either by sidestepping the framework altogether (vanilla LoRA [14] and Imbalance-Reg [40], which apply per-coordinate adaptive updates directly on the factors) or by taking $\mathcal{F}_t = \mathbf{I}$ with block-diagonal approximations of $(J_G^* J_G)$ (LoRA+ [11], Riemannian Preconditioned LoRA [37]). LoRA-Pro [33] stays in the affine solution set of (7) by minimizing the Frobenius residual $\|J_G[\Delta_{B_t}, \Delta_{A_t}] - \mathcal{F}_t^{-1} \mathbf{G}_t\|_F^2$; its AdamW variant pairs a non-trivial $\mathcal{F}_t$ on $\mathbf{W}$ with a Frobenius (rather than $\mathcal{F}_t$ -weighted) residual, mismatching the preconditioner’s metric, and explicitly maintains $\mathbf{W}$ -space first/second moments at $\mathcal{O}(mn)$ memory prohibitive at LLM scale. Manifold-based methods (Riemannian Muon [5], RAdamW [4]) take a Riemannian gradient step on $\mathcal{M}_r$ in the ambient $\mathbf{W}$ -space and rely on a retraction back to the manifold, rather than a closed-form solution of (7) in factor coordinates. A gradient-statistics-aware $J_G^* \mathcal{F}_t J_G$ paired with $\mathcal{O}((m+n)r)$ memory in the LoRA factor space remains an underexplored design point.

We target this point by observing that, even though $J_G^* \mathcal{F}_t J_G$ is singular, the linear system $J_G^* \mathcal{F}_t J_G [\mathbf{P}, \mathbf{Q}] = J_G^* (\mathbf{G}_t)$ on the factor pair $[\mathbf{P}, \mathbf{Q}]$ is always consistent: its solution set is a non-empty r^2 -dimensional affine subspace, the directions in factor space whose induced $\mathbf{W}$ -update equals $\mathcal{F}_t^{-1} \mathbf{G}_t$ projected onto $\mathbb{T}_t = \text{range}(J_G)$, the subspace of $\mathbf{W}$ -changes a single LoRA step can express. Designing a LoRA optimizer in this framework therefore decomposes into two coupled choices (§ 3): (i) which gradient-statistics-aware preconditioner $\mathcal{F}_t$ to use on $\mathbf{W}$, and (ii) how to select a particular element of the affine solution set. For (i), we adopt the Adafactor [28] diagonal Kronecker form $\mathcal{F}_t = \mathbf{L}_t \otimes \mathbf{R}_t$ (with operator square root $\mathcal{H}_t = \mathcal{F}_t^{1/2}$, acting as $\mathcal{H}_t \mathbf{Y} = \mathbf{L}_t^{1/2} \mathbf{Y} \mathbf{R}_t^{1/2}$), the cheapest non-trivial $\mathbf{W}$ -space preconditioner ($\mathcal{O}(m+n)$ memory). For (ii), we use the fact that all elements of the affine solution set induce the same $\mathbf{W}$ -update (the $\mathcal{H}_t$ -orthogonal projection of $\mathcal{H}_t^{-1} \mathbf{G}_t$ onto $\mathbb{T}_t$) but trace different factor trajectories, and pick the element that minimizes the $\mathcal{H}_t$ -norm imbalance $\|\Delta_{B_t} \mathbf{A}_t - \mathbf{B}_t \Delta_{A_t}\|_{\mathcal{H}_t}^2$ between the two factor contributions to the $\mathbf{W}$ -update. The resulting algorithm, **Adafactor Preconditioned Low-Rank Adaptation (AdaPreLoRA)**, admits a closed-form factor update at $\mathcal{O}(r^3)$ extra cost and keeps the optimizer state at $\mathcal{O}((m+n)r)$. By construction, its $\mathbf{W}$ -update is the closest point in $\mathbb{T}_t$ to the Adafactor-preconditioned direction $\mathcal{H}_t^{-1} \mathbf{G}_t$ under the $\mathcal{H}_t$ -weighted norm. Cheap factor-space schemes lack this guarantee, since their updates do not arise as a $\mathbf{W}$ -space preconditioned direction projected onto $\mathbb{T}_t$.

Empirically (§ 4), AdaPreLoRA matches or outperforms vanilla LoRA, Scaled AdamW, LoRA-Pro AdamW, and SOAP across GPT-2 (E2E), Mistral-7B and Qwen2-7B (GLUE, ARC, GSM8K), and diffusion-model fine-tuning, while matching Scaled AdamW’s peak memory and avoiding the $\sim 2\times$ memory overhead of LoRA-Pro AdamW. Our contributions are:

- A unified framework that recasts existing LoRA optimizers as instances of the consistent linear system $J_G^* \mathcal{F}_t J_G [\Delta_{B_t}, \Delta_{A_t}] = J_G^* (\mathbf{G}_t)$, parameterized by the choice of preconditioner $\mathcal{F}_t$ and the rule that selects an element of the affine solution set (§ 2.2).
- **AdaPreLoRA**, a LoRA optimizer whose $\mathbf{W}$ -update is the closest point in $\mathbb{T}_t$ to the Adafactor-preconditioned direction $\mathcal{H}_t^{-1} \mathbf{G}_t$ under the $\mathcal{H}_t$ -weighted norm, recovered in closed form at $\mathcal{O}((m+n)r)$ memory (§ 3).
- Experimental evidence that the resulting update direction is competitive with or improves over both cheap factor-space and pseudoinverse-based baselines, including at the 7B parameter scale.

2 Background and Related Work

This section sets up the LoRA optimization problem (§ 2.1), identifies the singular factor-space operator $J_G^* \mathcal{F}_t J_G$ that any $\mathbf{W}$ -space preconditioner $\mathcal{F}_t$ induces, unifies existing LoRA optimizers as different choices of $\mathcal{F}_t$ together with different generalized inverses of $J_G^* \mathcal{F}_t J_G$ (§ 2.2), and reviews adaptive preconditioner families on $\mathbf{W}$ (§ 2.3). Throughout the paper, calligraphic letters

$(\mathcal{F}_t, \mathcal{H}_t, \tilde{\mathcal{P}}_{\mathbb{T}_t}, J_{\mathcal{G}})$ denote linear operators on $\mathbb{R}^{m \times n}$, while bold letters $(\mathbf{L}_t, \mathbf{R}_t, \mathbf{B}_t, \mathbf{A}_t, \mathbf{G}_t)$ denote matrices; a complete notation table is given in Appendix A.

2.1 LoRA Setup and Its Singular Jacobian

As a representative parameter-efficient fine-tuning method, low-rank fine-tuning freezes the pretrained weight $\mathbf{W}_0 \in \mathbb{R}^{m \times n}$ and assumes that the weight update $\mathbf{W}$ admits a low-rank factorization $\mathbf{W} = \mathbf{B}\mathbf{A}$ with $\mathbf{B} \in \mathbb{R}^{m \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times n}$, and $r \ll \min\{m, n\}$ [14]. The fine-tuning objective is

$$\min_{\mathbf{B} \in \mathbb{R}^{m \times r}, \mathbf{A} \in \mathbb{R}^{r \times n}} \mathcal{L}(\mathbf{W}_0 + \mathcal{G}([\mathbf{B}, \mathbf{A}])), \quad \text{where } \mathcal{G}([\mathbf{B}, \mathbf{A}]) = \mathbf{B}\mathbf{A}.$$

Under this generator, the Jacobian operator $J_{\mathcal{G}}([\mathbf{B}_t, \mathbf{A}_t]) : [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}] \rightarrow \mathbb{R}^{m \times n}$ and its adjoint $J_{\mathcal{G}}^*([\mathbf{B}_t, \mathbf{A}_t]) : \mathbb{R}^{m \times n} \rightarrow [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$ act as

$$J_{\mathcal{G}}([\mathbf{B}_t, \mathbf{A}_t])[P, Q] = P\mathbf{A}_t + \mathbf{B}_t Q, \quad J_{\mathcal{G}}^*([\mathbf{B}_t, \mathbf{A}_t])(C) = [C\mathbf{A}_t^\top, \mathbf{B}_t^\top C], \quad (1)$$

on factor-space directions $[P, Q] \in [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$ and $\mathbf{W}$ -space directions $C \in \mathbb{R}^{m \times n}$, respectively. We abbreviate $J_{\mathcal{G}} := J_{\mathcal{G}}([\mathbf{B}_t, \mathbf{A}_t])$ and $J_{\mathcal{G}}^* := J_{\mathcal{G}}^*([\mathbf{B}_t, \mathbf{A}_t])$ when the base point is clear; detailed derivations appear in Proposition B.1. The chain rule gives the factor gradients $\mathbf{G}_{\mathbf{B}_t} = \mathbf{G}_t \mathbf{A}_t^\top$, $\mathbf{G}_{\mathbf{A}_t} = \mathbf{B}_t^\top \mathbf{G}_t$, with $\mathbf{G}_t = \nabla_{\mathbf{W}_t} \mathcal{L}(\mathbf{W}_0 + \mathbf{W}_t)$, equivalently

$$J_{\mathcal{G}}^*(\mathbf{G}_t) = [\mathbf{G}_{\mathbf{B}_t}, \mathbf{G}_{\mathbf{A}_t}]. \quad (2)$$

Thus $J_{\mathcal{G}}$ is the central operator linking factor-space and $\mathbf{W}$ -space updates, and its properties determine what factor-space optimizers can achieve.

Unfortunately, $J_{\mathcal{G}}$ has a non-trivial kernel: *the Jacobian $J_{\mathcal{G}}$ is rank-deficient*. The Jacobian formula in (1) immediately produces a family of factor-space directions that $J_{\mathcal{G}}$ maps to $\mathbf{0} \in \mathbb{R}^{m \times n}$: for any $\mathbf{X} \in \mathbb{R}^{r \times r}$,

$$J_{\mathcal{G}}[\mathbf{B}_t \mathbf{X}, -\mathbf{X} \mathbf{A}_t] = \mathbf{B}_t \mathbf{X} \mathbf{A}_t - \mathbf{B}_t \mathbf{X} \mathbf{A}_t = \mathbf{0}, \quad (3)$$

so $[\mathbf{B}_t \mathbf{X}, -\mathbf{X} \mathbf{A}_t] \in \ker(J_{\mathcal{G}})$. When $\mathbf{B}_t$ has column rank r and $\mathbf{A}_t$ has row rank r , this family is the entire kernel: $\ker(J_{\mathcal{G}}) = \{[\mathbf{B}_t \mathbf{X}, -\mathbf{X} \mathbf{A}_t] : \mathbf{X} \in \mathbb{R}^{r \times r}\}$, an r^2 -dimensional subspace, so $\text{rank}(J_{\mathcal{G}}) = (m+n)r - r^2$ (Proposition B.2, Appendix B.2).

This rank deficiency also constrains the form of preconditioners in factor space. Since practical preconditioners are typically built as approximations to the Fisher information in the optimized parameterization, the natural preconditioner for the factors $[\mathbf{B}_t, \mathbf{A}_t]$ is the empirical Fisher formed from the per-sample factor gradients $[\mathbf{G}_{\mathbf{B}_t}^{(i)}, \mathbf{G}_{\mathbf{A}_t}^{(i)}] := [\nabla_{\mathbf{B}} \mathcal{L}_i(\mathbf{W}_t), \nabla_{\mathbf{A}} \mathcal{L}_i(\mathbf{W}_t)]$. That is,

$$\mathcal{E}_t := \frac{1}{N} \sum_{i=1}^N \langle [\mathbf{G}_{\mathbf{B}_t}^{(i)}, \mathbf{G}_{\mathbf{A}_t}^{(i)}], \cdot \rangle [\mathbf{G}_{\mathbf{B}_t}^{(i)}, \mathbf{G}_{\mathbf{A}_t}^{(i)}]$$

Adaptive optimizers such as Adam [16], AdaFactor [28], Shampoo [10], and K-FAC [20] may be viewed as structured approximations to $\mathcal{E}_t$: diagonal, rank-1 Kronecker, full Kronecker, and layerwise Kronecker, respectively, with the explicit sum replaced in practice by a running average over mini-batches. However, as our experiments and prior work show, these optimizers often perform poorly in the factorized setting. This motivates a closer look at the structure of $\mathcal{E}_t$.

Because each $\mathcal{L}_i$ depends on the factors only through $\mathbf{W} = \mathbf{B}\mathbf{A}$, the chain rule gives $[\mathbf{G}_{\mathbf{B}_t}^{(i)}, \mathbf{G}_{\mathbf{A}_t}^{(i)}] = J_{\mathcal{G}}^*(\mathbf{G}_t^{(i)})$, where $\mathbf{G}_t^{(i)} := \nabla_{\mathbf{W}} \mathcal{L}_i(\mathbf{W}_t)$. Hence

$$\mathcal{E}_t = \frac{1}{N} \sum_{i=1}^N \langle J_{\mathcal{G}}^*(\mathbf{G}_t^{(i)}), \cdot \rangle J_{\mathcal{G}}^*(\mathbf{G}_t^{(i)}) = J_{\mathcal{G}}^* \left(\frac{1}{N} \sum_{i=1}^N \mathbf{G}_t^{(i)} \langle \mathbf{G}_t^{(i)}, \cdot \rangle \right) J_{\mathcal{G}} = J_{\mathcal{G}}^* \mathcal{F}_t J_{\mathcal{G}}, \quad (4)$$

where $\mathcal{F}_t := \frac{1}{N} \sum_{i=1}^N \mathbf{G}_t^{(i)} \langle \mathbf{G}_t^{(i)}, \cdot \rangle$ on the right is the empirical Fisher in $\mathbf{W}$ -space [17]. Equivalently, in matrix form,

$$\mathcal{F}_t = \frac{1}{N} \sum_{i=1}^N \text{vec}(\mathbf{G}_t^{(i)}) \text{vec}(\mathbf{G}_t^{(i)})^\top \in \mathbb{R}^{mn \times mn}. \quad (5)$$

Since J_G is rank-deficient, the pullback $J_G^* \mathcal{F}_t J_G$ is necessarily singular. When $\mathcal{F}_t \succ 0$, $\ker(J_G^* \mathcal{F}_t J_G) = \ker(J_G)$ is non-trivial, so $J_G^* \mathcal{F}_t J_G$ is singular for any choice of $\mathcal{F}_t$, and the corresponding preconditioned update on $[B_t, A_t]$,

$$[B_{t+1}, A_{t+1}] = [B_t, A_t] - \eta_t (J_G^* \mathcal{F}_t J_G)^{-1} [G_{B_t}, G_{A_t}], \quad (6)$$

is ill-defined.

Obstruction 1 (non-invertibility). The operator $J_G^* \mathcal{F}_t J_G$ is non-invertible for any choice of $\mathcal{F}_t$, so the update (6) is ill-defined.

One natural remedy is to replace the inverse with a generalized inverse, but different choices produce different factor updates, so the ill-definedness shifts from non-existence to non-uniqueness. A canonical choice is the Moore–Penrose pseudoinverse $(J_G^* \mathcal{F}_t J_G)^\dagger$:

$$[\Delta_{B_t}, \Delta_{A_t}] := (J_G^* \mathcal{F}_t J_G)^\dagger [G_{B_t}, G_{A_t}] \in [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}].$$

By property of the Moore–Penrose pseudoinverse, $[P, Q]$ is the unique minimum-Frobenius-norm element of the affine solution set

$$\{[P, Q] \in [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}] : J_G^* \mathcal{F}_t J_G [P, Q] = [G_{B_t}, G_{A_t}]\},$$

which is consistent since $\text{range}(J_G^*) = \text{range}(J_G^* \mathcal{F}_t J_G)$ as $\mathcal{F}_t \succ 0$, and has dimension $\dim \ker(J_G^* \mathcal{F}_t J_G) = \dim \ker(J_G) = r^2$. Other generalized inverses of $J_G^* \mathcal{F}_t J_G$ correspond to different elements of this affine solution set, differing by an element of $\ker(J_G)$.

Obstruction 2 (non-uniqueness of generalized inverse). Generalized inverses of $J_G^* \mathcal{F}_t J_G$ are not unique; the resulting factor updates from different generalized inverses can differ by any element of $\ker(J_G)$.

Existing LoRA optimizers in § 2.2 differ in the choice of $\mathcal{F}_t$ on W and in the rule that selects an element of this affine solution set; § 2.3 reviews the standard families of $\mathcal{F}_t$. Our method (§ 3) instantiates this framework with the Adafactor diagonal Kronecker and an $\mathcal{H}_t$ -balance criterion that selects a unique element of the affine solution set.

2.2 Existing LoRA Optimizers

Although $J_G^* \mathcal{F}_t J_G$ is singular, the linear system

$$J_G^* \mathcal{F}_t J_G [\Delta_{B_t}, \Delta_{A_t}] = J_G^* (G_t) \quad (7)$$

in the factor update $[\Delta_{B_t}, \Delta_{A_t}]$ is consistent, with RHS equal to the factor-gradient pair $[G_{B_t}, G_{A_t}]$ by (2). We organize existing LoRA optimizers by (i) which invertible surrogate for $J_G^* \mathcal{F}_t J_G$ they use and (ii) the choice of $\mathcal{F}_t$ on W . Table 1 summarizes the resulting design space, and we walk through the main families below.

Vanilla LoRA / Imbalance-Reg / LoRA-RITE (diagonal approximation of $\mathcal{E}_t$, ignoring $J_G^* \mathcal{F}_t J_G$). Vanilla LoRA [14] ignores $J_G^* \mathcal{F}_t J_G$ and approximates the empirical Fisher $\mathcal{E}_t$ on the factor space by a per-coordinate diagonal estimated directly from the factor gradients $[G_{B_t}, G_{A_t}]$ via AdamW. Imbalance-Regularized LoRA [40] keeps the same diagonal Fisher estimate and adds a penalty $\|B_t^\top B_t - A_t A_t^\top\|_F^2$ to align the factor spectra. LoRA-RITE [36] replaces the diagonal estimate with a matrix-form second moment $V_t \in \mathbb{R}^{r \times r}$ accumulated on the polar/QR-reparameterized factor gradients, yielding a transformation-invariant factor-space update at $\mathcal{O}(r^2)$ extra memory.

LoRA+ [11] and Riemannian Preconditioned LoRA [37] (block-diagonal surrogates for $J_G^* J_G$). Specializing (7) to $\mathcal{F}_t = I$, the operator $J_G^* J_G$ (Proposition B.1) decomposes into block-diagonal and cross terms. LoRA+ approximates $J_G^* J_G$ by the block-scaling identity surrogate $\text{diag}(I_m, \lambda I_n)$ for a fixed scalar $\lambda > 0$ and inverts it, giving asymmetric per-block scalar rescaling between the B_t and A_t updates. Riemannian Preconditioned LoRA approximates $J_G^* J_G$ by its block-diagonal part $\text{diag}(A_t A_t^\top, B_t^\top B_t)$ and inverts it, yielding the explicit factor update $\Delta_{B_t} = G_{B_t} (A_t A_t^\top)^{-1}$, $\Delta_{A_t} = (B_t^\top B_t)^{-1} G_{A_t}$, well-defined whenever B_t, A_t have full rank. Both updates differ from any element of the affine solution set of (7).

LoRA-Pro [33]. LoRA-Pro solves a different system from (7): it minimizes the Frobenius residual $\|J_G [\Delta_{B_t}, \Delta_{A_t}] - \mathcal{F}_t^{-1} G_t\|_F^2$, whose normal equations $J_G^* J_G [\Delta_{B_t}, \Delta_{A_t}] = J_G^* (\mathcal{F}_t^{-1} G_t)$ coincide

Table 1: Existing LoRA optimizers as instances of the framework (7), grouped by how they handle the singular operator $J_G^* \mathcal{F}_t J_G$. Inversion strategy = how $(J_G^* \mathcal{F}_t J_G)^{-1}$ is replaced or approximated; $\mathcal{F}_t$ = the gradient-statistics structure on $\mathbf{W}$ used (“—” means the method bypasses (7) and does not instantiate $\mathcal{F}_t$). Per-step cost and memory are per-layer beyond forward/backward through $\mathbf{BA}$ and storing $[\mathbf{B}_t, \mathbf{A}_t]$. $\mathbf{g} := \text{vec}(\mathbf{G}_t) \in \mathbb{R}^{mn}$ denotes the vectorized $\mathbf{W}$ -space gradient.

Method	Inversion strategy	$\mathcal{F}_t$ on $\mathbf{W}$	Per-step cost	Memory
<i>Bypass (7) (factor-space AdamW)</i>				
Vanilla LoRA [14]	—	—	$\mathcal{O}((m+n)r)$	$\mathcal{O}((m+n)r)$
LoRA-RITE [36]	—	—	$\mathcal{O}((m+n)r^2)$	$\mathcal{O}((m+n)r)$
<i>Block-diagonal approx. of $J_G^* J_G$</i>				
LoRA+ [11]	Block-scaling	$\mathbf{I}$	$\mathcal{O}((m+n)r)$	$\mathcal{O}((m+n)r)$
Riem. Precond. [37]	Block-diag	$\mathbf{I}$	$\mathcal{O}((m+n)r^2)$	$\mathcal{O}((m+n)r)$
<i>Pseudoinverse of J_G</i>				
LoRA-Pro SGD [33]	$J_G^\dagger \mathcal{F}_t^{-1}$	$\mathbf{I}$	$\mathcal{O}((m+n)r^2)$	N/A
LoRA-Pro AdamW [33]	$J_G^\dagger \mathcal{F}_t^{-1}$	$\text{diag}(\mathbf{g} \odot \mathbf{g})^{\frac{1}{2}}$	$\mathcal{O}(mnr)$	$\mathcal{O}(mn)$
<i>Riemannian gradient descent on $\mathcal{M}_r$</i>				
Riem. Muon [5]	RGD on $\mathcal{M}_r$	$(\mathbf{G}_t \mathbf{G}_t^\top)^{1/2}$	$\mathcal{O}((m+n)r^2)$	$\mathcal{O}((m+n)r)$
RAdamW [4]	RGD on $\mathcal{M}_r$	$\text{diag}(\mathbf{L}_{\text{Sh}} \otimes \mathbf{R}_{\text{Sh}})^{\frac{1}{4}}$	$\mathcal{O}((m+n)r^2)$	$\mathcal{O}((m+n)r)$
AdaPreLoRA (ours)	solve (7)	Adafactor diag-Kron	$\mathcal{O}((m+n)r^2)$	$\mathcal{O}((m+n)r)$

with (7) only when $\mathcal{F}_t = \mathbf{I}$. Its AdamW variant pairs a non-trivial $\mathcal{F}_t$ on $\mathbf{W}$ with the Frobenius (rather than $\mathcal{F}_t$ -weighted) residual, mismatching the preconditioner’s metric, and explicitly maintains $\mathbf{W}$ -space first/second moments at $\mathcal{O}(mn)$ memory prohibitive at LLM scale. In contrast, our (11) measures the residual under the $\mathcal{F}_t$ -induced $\mathcal{H}_t$ -norm consistent with the preconditioner.

Manifold-based methods on $\mathcal{M}_r$. Rather than solving (7) on the factor space, this line of work performs Riemannian gradient descent on the rank- r matrix manifold $\mathcal{M}_r$. Riemannian Muon [5] uses retraction-based Muon updates on $\mathcal{M}_r$, applying Muon orthogonalization (replacing all singular values by 1) on the tangent space; the resulting step is equivalent to a per-step spectral $\mathbf{W}$ -space preconditioner $\mathcal{F}_t = (\mathbf{G}_t \mathbf{G}_t^\top)^{1/2}$ (no accumulation across steps). RAdaGrad / RAdamW [4] run Riemannian gradient descent on $\mathcal{M}_r$ under a Shampoo $\mathbf{W}$ -space preconditioner $\mathcal{F}_t = (\mathbf{L}_{\text{Sh}} \otimes \mathbf{R}_{\text{Sh}})^{\frac{1}{4}}$ restricted to the manifold tangent space, achieving a similar $\mathcal{F}_t$ -aware behaviour to ours but via a retraction step on $\mathcal{M}_r$ instead of a closed-form solution of (7) in factor coordinates.

Other directions (LoRA-RITE / LoRA-GA). LoRA-RITE [36] introduces transformation invariance via a polar-decomposition-based reparameterization of the factor coordinates, with $\mathcal{F}_t = \mathbf{I}$; LoRA-GA [32] addresses initialization through spectral alignment with full fine-tuning gradients.

These methods reveal a recurring trade-off: cheap factor-space schemes (identity replacement, block-diagonal approximations) typically take $\mathcal{F}_t = \mathbf{I}$ and discard gradient statistics, while methods admitting a non-trivial $\mathcal{F}_t$ (LoRA-Pro AdamW) pay $\mathcal{O}(mn)$ memory or operate in the ambient $\mathbf{W}$ -space. A gradient-statistics-aware $\mathcal{F}_t$ paired with $\mathcal{O}((m+n)r)$ memory in the LoRA factor space remains an underexplored design point, which our method (§ 3) targets via the Adafactor diagonal Kronecker $\mathcal{F}_t = \mathbf{L}_t \otimes \mathbf{R}_t$ together with a closed-form solution of (7) that picks a specific element of the r^2 -dimensional affine solution set.

2.3 Choosing $\mathcal{F}_t$: Adaptive Preconditioner Toolkit on $\mathbf{W}$

The gap identified above asks for an $\mathcal{F}_t$ that is gradient-statistics-based yet cheap on $\mathbf{W}$. We review the standard families of $\mathbf{W}$ -space preconditioners, organized by memory cost. All families construct a second-moment-based preconditioner $\mathcal{F}_t : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ from gradient outer-product statistics of the form $\mathbf{G}_t \mathbf{G}_t^\top$ or $\mathbf{G}_t \odot \mathbf{G}_t$, and produce the preconditioned update $\mathbf{W}_{t+1} = \mathbf{W}_t - \eta_t \mathcal{F}_t^{-1} \mathbf{G}_t$; they differ in the structure imposed on $\mathcal{F}_t$, which trades off expressiveness against cost. AdaGrad [8] and Adam [16] approximate $\mathcal{F}_t$ by its diagonal $\mathbf{h}_t \in \mathbb{R}^{mn}$ as an exponential moving average of

$\mathbf{G}_t \odot \mathbf{G}_t$, yielding per-coordinate rescaling that ignores the matrix structure of $\mathbf{G}_t$ at memory $\mathcal{O}(mn)$. Adafactor [28] compresses this further into a rank-1 Kronecker form by maintaining only the row sums $\mathbf{l}_t \in \mathbb{R}^m$ and column sums $\mathbf{r}_t \in \mathbb{R}^n$ of $\mathbf{G}_t \odot \mathbf{G}_t$ (the elementwise Hadamard product), dropping the memory cost to $\mathcal{O}(m+n)$. Shampoo [10] maintains $\mathbf{L}_{\text{Sh},t} = \mathbf{L}_{\text{Sh},t-1} + \mathbf{G}_t \mathbf{G}_t^\top \in \mathbb{R}^{m \times m}$ and $\mathbf{R}_{\text{Sh},t} = \mathbf{R}_{\text{Sh},t-1} + \mathbf{G}_t^\top \mathbf{G}_t \in \mathbb{R}^{n \times n}$ and updates by $\mathbf{W}_{t+1} = \mathbf{W}_t - \mathbf{L}_{\text{Sh},t}^{-\frac{1}{4}} \mathbf{G}_t \mathbf{R}_{\text{Sh},t}^{-\frac{1}{4}}$; SOAP [22, 30] runs Adam in the eigenbasis of the Shampoo preconditioner; and K-FAC [19, 20] factorizes $\mathcal{F}_t$ as the Kronecker product of activation and gradient covariances. All three impose $\mathcal{O}(m^2 + n^2)$ memory and $\mathcal{O}(m^3 + n^3)$ per-step inverse cost, which dominates LoRA’s budgets. Among these candidates, the Adafactor diagonal Kronecker form $\mathcal{F}_t = \mathbf{L}_t \otimes \mathbf{R}_t$ is the only one that is simultaneously gradient-statistics-based and cheap ($\mathcal{O}(m+n)$ memory). Our method (§ 3) adopts this candidate and pairs it with a closed-form solution of the linear system (7) that respects the LoRA factorization.

3 The Proposed Algorithms

We instantiate the framework (7) of § 2.2 with two specific choices: (i) for the $\mathbf{W}$ -space preconditioner, the Adafactor diagonal Kronecker form $\mathcal{H}_t = \mathbf{L}_t^{1/2} \otimes \mathbf{R}_t^{1/2}$ on $\mathbf{W}$ (§ 3.1); (ii) for the element of the affine solution set of (7), the unique minimizer of the $\mathcal{H}_t$ -imbalance criterion (Solution 2). Choice (i) avoids inverting the singular operator $J_{\mathcal{G}}^* \mathcal{H}_t J_{\mathcal{G}}$ directly (Obstruction 1); choice (ii) resolves the r^2 -dimensional ambiguity over $\ker(J_{\mathcal{G}})$ (Obstruction 2). The closed-form factor update is given in Theorem 3.2. Figure 1 contrasts the resulting $\mathbf{W}$ -update geometry against LoRA-Pro and Riemannian Preconditioned LoRA [37] under the $\mathcal{H}_t$ -weighted inner product.

3.1 The Adafactor Preconditioner $\mathcal{H}_t = \mathbf{L}_t^{1/2} \otimes \mathbf{R}_t^{1/2}$

We adopt the diagonal Kronecker preconditioner $\mathcal{H}_t = \mathbf{L}_t^{1/2} \otimes \mathbf{R}_t^{1/2}$ on $\mathbf{W}$, where $\mathbf{L}_t, \mathbf{R}_t$ are the Adafactor [28] rank-1 second-moment estimate of $\mathbf{G}_t \odot \mathbf{G}_t$:

$$\begin{aligned} \mathbf{L}_t &= \text{diag}(\mathbf{l}_t / \sqrt{\|\mathbf{l}_t\|_1}), \quad \mathbf{R}_t = \text{diag}(\mathbf{r}_t / \sqrt{\|\mathbf{r}_t\|_1}), \quad \text{with} \\ \mathbf{l}_t &= \beta_1 \mathbf{l}_{t-1} + (1 - \beta_1) \sum_{j=1}^n (\mathbf{G}_t \odot \mathbf{G}_t)_{i,j}, \quad \mathbf{r}_t = \beta_2 \mathbf{r}_{t-1} + (1 - \beta_2) \sum_{i=1}^m (\mathbf{G}_t \odot \mathbf{G}_t)_{i,j}, \end{aligned} \quad (8)$$

where $\odot$ denotes the Hadamard product, $\|\cdot\|_1$ denotes the ℓ_1 -norm, and $\beta_1, \beta_2 \in [0, 1]$ are decay rates. The vectors $\mathbf{l}_t \in \mathbb{R}^m$ and $\mathbf{r}_t \in \mathbb{R}^n$ are the diagonals of the moving averages of $\mathbf{G}_t \mathbf{G}_t^\top$ and $\mathbf{G}_t^\top \mathbf{G}_t$, respectively, so $\mathbf{l}_t \mathbf{r}_t^\top$ is the rank-1 Adafactor approximation of $\mathbf{G}_t \odot \mathbf{G}_t$ [28]. The memory cost is $\mathcal{O}(m+n)$.

We treat $\mathcal{H}_t$ as an operator on $\mathbb{R}^{m \times n}$, defined by $\mathcal{H}_t \mathbf{Y} := \mathbf{L}_t^{1/2} \mathbf{Y} \mathbf{R}_t^{1/2}$ for any $\mathbf{Y} \in \mathbb{R}^{m \times n}$, with inverse $\mathcal{H}_t^{-1} \mathbf{K} = \mathbf{L}_t^{-1/2} \mathbf{K} \mathbf{R}_t^{-1/2}$ (so $\mathcal{H}_t = \mathcal{F}_t^{1/2}$ for the underlying second-moment operator $\mathcal{F}_t = \mathbf{L}_t \otimes \mathbf{R}_t$). The $1/2$ -power form ensures that the resulting preconditioned direction $\mathcal{H}_t^{-1} \mathbf{G}_t = \mathbf{L}_t^{-1/2} \mathbf{G}_t \mathbf{R}_t^{-1/2}$ matches Adafactor’s standard square root second-moment update rule [28] and the $1/2$ -power Shampoo preconditioner advocated by SOAP [22, 30] as the Frobenius-optimal Kronecker approximation of the gradient outer-product matrix $\sum_t \mathbf{G}_t \mathbf{G}_t^\top$. The associated inner product on $\mathbb{R}^{m \times n}$ is

$$\langle \mathbf{Y}, \mathbf{Z} \rangle_{\mathcal{H}_t} := \langle \mathcal{H}_t \mathbf{Y}, \mathbf{Z} \rangle = \langle \mathbf{L}_t^{1/2} \mathbf{Y} \mathbf{R}_t^{1/2}, \mathbf{Z} \rangle, \quad (9)$$

where $\langle \cdot, \cdot \rangle$ is the Frobenius inner product.

3.2 Solving the Linear System on Factor Space

With $\mathcal{H}_t = \mathbf{L}_t^{1/2} \otimes \mathbf{R}_t^{1/2}$ from § 3.1, the factor-space linear system (7) becomes

$$J_{\mathcal{G}}^* \mathcal{H}_t J_{\mathcal{G}} [\Delta_{\mathbf{B}_t}, \Delta_{\mathbf{A}_t}] = J_{\mathcal{G}}^* (\mathbf{G}_t), \quad (10)$$

in the candidate factor update $[\Delta_{\mathbf{B}_t}, \Delta_{\mathbf{A}_t}] \in \mathbb{R}^{m \times r} \times \mathbb{R}^{r \times n}$. The operator $J_{\mathcal{G}}^* \mathcal{H}_t J_{\mathcal{G}}$ is singular (Obstruction 1), so we cannot invert it.

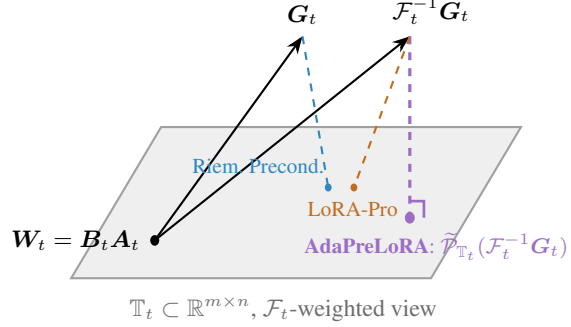
Solution 1 (Bypass Obstruction 1: solve the equivalent least-squares problem). Equation (10) is the normal equation of

$$\min_{\Delta_{B_t}, \Delta_{A_t}} \|J_G[\Delta_{B_t}, \Delta_{A_t}] - \mathcal{H}_t^{-1} G_t\|_{\mathcal{H}_t}^2, \quad (11)$$

so solving (11) replaces inverting $J_G^* \mathcal{H}_t J_G$.

The following theorem characterizes the solution set of (11).

Figure 1: Geometric contrast of LoRA optimizers under the $\mathcal{F}_t$ -weighted inner product on $\mathbb{R}^{m \times n}$; $\mathcal{F}_t^{-1} G_t$ is the gradient under this inner product, and all updates land in $\mathbb{T}_t = \text{range}(J_G)$. From $\mathcal{F}_t^{-1} G_t$, **AdaPreLoRA** drops $\mathcal{F}_t$ -orthogonally; **LoRA-Pro** does not realize an orthogonal projection under the $\mathcal{F}_t$ -weighted inner product. **Riem. Precond.** lies in $\mathbb{T}_t$ but is neither a Frobenius nor an $\mathcal{F}_t$ -weighted orthogonal projection. LoRA-Pro and AdaPreLoRA coincide when $\mathcal{F}_t = I$.



Theorem 3.1 (Solution set of (11)). Let $\tilde{G}_t := \mathcal{H}_t^{-1} G_t$. Since $J_G[\Delta_{B_t}, \Delta_{A_t}] \in \mathbb{T}_t = \text{range}(J_G)$, the minimum of (11) is attained iff

$$\Delta_{B_t} A_t + B_t \Delta_{A_t} = \tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{G}_t), \quad (12)$$

the $\mathcal{H}_t$ -orthogonal projection of $\tilde{G}_t$ onto $\mathbb{T}_t$ (closed form in Appendix B.5). The minimizers form an r^2 -parameter family (Appendix B.5, Lemma B.1)

$$\begin{aligned} \Delta_{B_t}(X_t) &= [L_t^{-1/2} - B_t(B_t^\top L_t^{1/2} B_t)^{-1} B_t^\top] G_{B_t}(A_t R_t^{1/2} A_t^\top)^{-1} + B_t X_t, \\ \Delta_{A_t}(X_t) &= (B_t^\top L_t^{1/2} B_t)^{-1} G_{A_t} R_t^{-1/2} - X_t A_t, \end{aligned} \quad X_t \in \mathbb{R}^{r \times r}, \quad (13)$$

where the offsets $(B_t X_t, -X_t A_t)$ parameterize $\ker(J_G)$.

By Theorem 3.1, every factor pair in (13) induces the common W -update $\tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{G}_t)$. The following solution selects a specific X_t to resolve this r^2 -dimensional ambiguity (Obstruction 2).

Solution 2 ($\mathcal{H}_t$ -balance fixes the $\ker(J_G)$ ambiguity). Among the X_t -family (13), we select the unique element by choosing X_t to minimize the $\mathcal{H}_t$ -imbalance $\|\Delta_{B_t} A_t - B_t \Delta_{A_t}\|_{\mathcal{H}_t}^2$ between the two factor contributions to the W -update.

This criterion balances the magnitudes of the two factor contributions, in the same spirit as the regularizer in Imbalance-Regularized LoRA [40] and the standard balance term used in nonconvex low-rank matrix recovery [29]. Combining Theorem 3.1 with Solutions 1 and 2 fixes X_t in closed form and yields the full AdaPreLoRA update.

Theorem 3.2 (AdaPreLoRA closed-form factor update). The unique factor update solving (11) together with the $\mathcal{H}_t$ -balance criterion (Solution 2) is

$$\Delta_{B_t}^{\text{opt}} = (I - \frac{1}{2} \tilde{P}_{B_t}) L_t^{-1/2} G_{B_t} (A_t R_t^{1/2} A_t^\top)^{-1}, \quad \Delta_{A_t}^{\text{opt}} = (B_t^\top L_t^{1/2} B_t)^{-1} G_{A_t} R_t^{-1/2} (I - \frac{1}{2} \tilde{Q}_{A_t}),$$

where the $\mathcal{H}_t$ -weighted projector matrices are

$$\tilde{P}_{B_t} := B_t (B_t^\top L_t^{1/2} B_t)^{-1} B_t^\top L_t^{1/2}, \quad \tilde{Q}_{A_t} := R_t^{1/2} A_t^\top (A_t R_t^{1/2} A_t^\top)^{-1} A_t. \quad (14)$$

Under the $\mathcal{H}_t$ -balance criterion (Solution 2), two features are worth highlighting: (i) the update depends on the gradient only through the low-rank factor gradients G_{A_t}, G_{B_t} , keeping per-step memory at $\mathcal{O}((m+n)r)$; (ii) the $\frac{1}{2}$ coefficients on the projectors are the signature of the $\mathcal{H}_t$ -balance choice. The full procedure, which we refer to as AdaPreLoRA throughout, is summarized as Algorithm 1 in the appendix; its computational complexity is analyzed in Appendix D. For practical use we also provide an Adam variant, given as Algorithm 2.

Table 2: GPT-2 fine-tuning on E2E at $r = 4$, across model size (small / medium). Bold/underline = best/second-best per metric per (model, optimizer family). Cross-rank ablation ($r \in \{16, 64\}$) and DART results are reported in Appendix E.

Model	r	Method	E2E				
			BLEU	NIST	MET	ROUGE-L	CIDEr
GPT-2 small	4	SGD	54.8	4.56	34.0	63.3	1.29
		Scaled GD	<u>68.5</u>	<u>8.72</u>	<u>45.5</u>	69.4	2.40
		LoRA-Pro SGD	68.4	<u>8.72</u>	<u>45.5</u>	<u>69.6</u>	<u>2.43</u>
		AdaPreLoRA SGD (ours)	69.5	8.77	46.5	71.5	2.50
		AdamW	69.1	8.75	46.0	70.5	2.47
		Scaled AdamW	<u>69.5</u>	<u>8.80</u>	<u>46.2</u>	<u>70.9</u>	<u>2.48</u>
		LoRA-Pro AdamW	69.2	8.73	45.9	70.8	2.47
		AdaPreLoRA AdamW (ours)	70.0	8.84	46.3	71.3	2.50
GPT-2 medium	4	SGD	66.6	8.54	44.2	68.2	2.32
		Scaled GD	69.2	8.71	46.3	70.9	2.48
		LoRA-Pro SGD	<u>69.7</u>	<u>8.77</u>	<u>46.5</u>	<u>70.9</u>	<u>2.50</u>
		AdaPreLoRA SGD (ours)	70.3	8.84	46.9	71.7	2.54
		AdamW	68.9	8.69	46.5	71.3	2.51
		Scaled AdamW	69.6	8.77	<u>46.6</u>	71.8	<u>2.52</u>
		LoRA-Pro AdamW	69.8	8.78	46.5	71.7	<u>2.52</u>
		AdaPreLoRA AdamW (ours)	70.3	8.84	46.7	71.8	2.53

4 Experimental Results

We evaluate AdaPreLoRA against representatives of the design families identified in Table 1: vanilla LoRA / AdamW (identity replacement), Scaled GD / Scaled AdamW (Riemannian Preconditioned LoRA [37] with SGD / AdamW; block-diagonal $J_G^* J_G$), LoRA-Pro SGD / AdamW [33] (Moore–Penrose $J_G^\dagger$), and SOAP [30] (direct on $\mathbf{W}$). Three axes test the trade-off identified in § 2.2: model scale (124M–355M GPT-2 vs. 7B Mistral/Qwen2), task family (NLU, reasoning, math, generation, image), and resource cost (peak GPU memory and per-step time). Following [33, 37], learning rates are independently tuned per optimizer via grid search; full hyperparameters are in Appendix E.1. All runs use PyTorch [25] on NVIDIA A100 GPUs.

4.1 Controlled study: GPT-2

We start with controlled fine-tuning of GPT-2 [26] (small, 124M; medium, 355M) on the E2E natural language generation challenge [24], sweeping rank $r \in \{4, 16, 64\}$ to probe the conditioning-vs. overparameterization trade-off and isolate the effect of $\mathcal{F}_t$ at small scale.

Table 2 reports E2E scores at $r = 4$ on both GPT-2 small and medium. AdaPreLoRA achieves the best or tied-best score on every metric across both SGD-based and AdamW-based families and both model sizes; the gain is largest in the SGD-based group, where vanilla LoRA’s identity replacement is most exposed, and persists for AdamW-based methods despite the smaller absolute headroom. Adding gradient statistics through Scaled GD or LoRA-Pro narrows but does not close this gap: AdaPreLoRA’s $\mathcal{H}_t$ -orthogonal projection of the Adafactor-preconditioned direction $\mathcal{H}_t^{-1} \mathbf{G}_t$ exploits curvature information that block-diagonal and Euclidean-projection schemes leave on the table. To further validate the effectiveness of AdaPreLoRA, we also conducted GPT-2 fine-tuning experiments on the DART [23] dataset, with the results reported in Table 3, the results show that the gains transfer to a different generation benchmark. Cross-rank ablations at $r \in \{16, 64\}$ (Appendix E.2, Table 10) confirm the same ordering.

4.2 Extension to 7B-scale LLMs: Mistral-7B and Qwen2-7B

We next test whether the controlled-setting gains carry over to the 7B parameter scale, using Mistral-7B [15] and Qwen2-7B [35]. For Mistral-7B we fine-tune on the GLUE [31] tasks RTE, CoLA, and

Table 3: Scores of GPT-2 small model (rank=4) fine-tuned using different optimizers. Evaluation is conducted on DART dataset.

Methods	BLEU↑	METEOR↑	chrF++↑	TER↓	BLEURT↑
SGD	41.2	0.63	0.59	0.52	0.33
Scaled GD	43.8	0.66	0.61	0.50	0.38
LoRA-Pro SGD	44.1	0.66	0.61	0.50	0.38
AdaPreLoRA SGD (ours)	44.6	0.66	0.62	0.49	0.39
AdamW	43.9	0.66	0.60	0.50	0.38
Scaled AdamW	44.8	0.67	0.62	0.49	0.40
LoRA-Pro AdamW	44.9	0.66	0.62	0.50	0.39
AdaPreLoRA AdamW (ours)	45.4	0.67	0.60	0.49	0.40

Table 4: Mistral-7B and Qwen2-7B fine-tuning accuracy (%) with rank $r = 8$. Bold/underline = best/second-best.

Method	Mistral-7B			Qwen2-7B			
	RTE	CoLA	MRPC	RTE	MRPC	ARC	GSM8K
AdamW	<u>89.4</u>	69.4	89.5	<u>90.6</u>	<u>90.0</u>	84.3	75.1
Scaled AdamW	89.1	71.5	<u>89.7</u>	<u>90.6</u>	87.0	<u>85.3</u>	74.2
LoRA-Pro AdamW	88.8	68.5	84.8	88.1	89.2	80.9	<u>75.7</u>
SOAP	88.9	68.3	86.3	86.6	89.2	80.9	73.2
AdaPreLoRA AdamW (ours)	89.5	<u>71.4</u>	90.0	91.0	90.4	85.6	76.4

MRPC, with three random seeds on RTE; for Qwen2-7B we additionally evaluate on the reasoning benchmark ARC [6] and the math benchmark GSM8K [7].

Table 5: Per-step GPU time and peak GPU memory on Mistral-7B. AdaPreLoRA matches the LoRA-level memory footprint of vanilla AdamW, while LoRA-Pro AdamW pays roughly $2\times$ peak memory for its full-weight gradient and moments.

Method	s/step	Mem (GB)
AdamW	0.24	25.8
Scaled AdamW	0.36	26.0
SOAP	0.65	26.4
LoRA-Pro AdamW	1.35	50.4
AdaPreLoRA SGD	0.46	21.5
AdaPreLoRA AdamW	0.99	26.0

Table 6: CLIP and FID scores on Mix-of-Show diffusion fine-tuning. Bold/underline = best/second-best.

Method	scaling=0.7		scaling=1	
	CLIP↑	FID↓	CLIP↑	FID↓
SGD	27.79	69.90	<u>31.40</u>	40.95
Scaled GD	<u>31.23</u>	35.86	30.60	29.62
LoRA-Pro SGD	31.47	34.30	30.48	29.19
AdaPreLoRA SGD	31.47	30.17	31.58	28.18
AdamW	31.47	34.15	30.68	27.80
Scaled AdamW	24.21	48.23	24.51	34.18
LoRA-Pro AdamW	<u>31.04</u>	<u>29.18</u>	30.60	28.18
AdaPreLoRA AdamW	31.47	29.01	30.73	27.13

Table 4 shows that AdaPreLoRA achieves the best accuracy on six of the seven settings (second-best on Mistral-7B CoLA, 0.1 point behind Scaled AdamW), and the lowest variance on Mistral-7B RTE among methods reporting std. Notably, gradient-statistics-aware baselines that pay $\mathcal{O}(mn)$ memory, LoRA-Pro AdamW (full-weight gradient + AdamW moments) and SOAP (full Shampoo on W) do not translate that extra cost into accuracy at 7B; both trail vanilla AdamW on multiple Qwen2-7B tasks (Qwen2-7B RTE: 88.1, 86.6 vs. AdamW 90.6). Table 5 contrasts per-step GPU time and peak memory on Mistral-7B GLUE-RTE: AdaPreLoRA-AdamW matches Scaled AdamW’s peak memory (26.0 GB) while LoRA-Pro AdamW requires 50.4 GB ($\sim 2\times$) due to materializing the full-weight gradient and its first/second moments; AdaPreLoRA-SGD has the lowest peak memory of all methods (21.5 GB).

4.3 Diffusion model personalization (Mix-of-Show)

To test transfer beyond NLP, we evaluate AdaPreLoRA on diffusion-model personalization with the Mix-of-Show framework [9], which uses Embedding Decomposed LoRA (EDLoRA) on the text encoder and U-Net of a Stable Diffusion backbone. Setup follows [9, 37] with embedding tuning disabled. We report CLIP score [12] (alignment with prompt; higher is better) and FID [13] (distributional similarity to reference images; lower is better) at LoRA scaling factors 0.7 and 1.0.

Table 6 shows that AdaPreLoRA achieves the lowest FID at every scaling-optimizer combination and the best CLIP at scaling 1.0 in both SGD- and AdamW-based families, while remaining competitive with the best baseline at scaling 0.7. This confirms that the gains from gradient-statistics-aware preconditioning carry over from text generation to image generation. Qualitative samples (Harry Potter / Hermione Granger) and per-prompt grids across LoRA scaling factors are in Appendix F.

5 Conclusion

We organized existing LoRA optimizers along two axes: (i) which invertible surrogate is used for the singular operator $J_{\mathcal{G}}^* \mathcal{F}_t J_{\mathcal{G}}$, and (ii) the choice of $\mathbf{W}$ -space preconditioner $\mathcal{F}_t$ (§ 2.2, Table 1). This framework exposes a previously underexplored design point: pairing a non-trivial gradient-statistics-aware $\mathcal{F}_t$ with LoRA’s $\mathcal{O}((m+n)r)$ memory budget. We instantiate this point as **AdaPreLoRA**, combining an Adafactor diagonal Kronecker preconditioner $\mathcal{H}_t$ on $\mathbf{W}$ with an $\mathcal{H}_t$ -balance criterion that selects a unique factor update from the affine solution set of (7); by construction, the resulting factor update is the closest LoRA approximation to the preconditioned $\mathbf{W}$ -space direction under the $\mathcal{H}_t$ -weighted norm, and admits a closed-form expression. Across GPT-2, Mistral-7B, Qwen2-7B, and Mix-of-Show diffusion personalization, AdaPreLoRA is competitive with or improves over a representative set of LoRA optimizers while keeping peak GPU memory at the LoRA optimizer level.

The two-axis framework is not specific to AdaPreLoRA: alternative choices for either axis remain open. Two natural extensions of AdaPreLoRA itself are (i) Mixture-of-Experts adapters, where each expert carries its own pair of low-rank factors and the framework applies per expert, and (ii) quantized backbones (QLoRA), where the $\mathbf{W}$ -space gradient must be reconstructed from quantized weights and dequantization-aware preconditioner statistics. Beyond language models, extending AdaPreLoRA to diffusion transformers (DiT) requires handling the cross-attention adapters and time-conditioning structure, where the assumption that a single $\mathcal{H}_t$ summarizes per-step gradient statistics may need to be relaxed. We leave a systematic study of these directions to future work.

References

- [1] P-A Absil, Robert Mahony, and Rodolphe Sepulchre. Optimization algorithms on matrix manifolds. In *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2009.
- [2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [3] Fengmiao Bian, Jian-Feng Cai, and Rui Zhang. A preconditioned riemannian gradient descent algorithm for low-rank matrix recovery. *SIAM Journal on Matrix Analysis and Applications*, 45(4):2075–2103, 2024.
- [4] Fengmiao Bian, Jinyang ZHENG, Ziyun Liu, Jianzhou Luo, and Jian-Feng Cai. Finding low-rank matrix weights in DNNs via riemannian optimization: RAdagrad and RAdamw. In *Advances in neural information processing systems (NeurIPS)*, 2026. URL <https://openreview.net/forum?id=tiGFicrmKm>.
- [5] Vladimir Bogachev, Vladimir Aletov, Alexander Molozhavenko, Denis Bobkov, Vera Soboleva, Aibek Alanov, and Maxim Rakhuba. Lora meets riemannion: Muon optimizer for parametrization-independent low-rank adapters. *arXiv preprint arXiv:2507.12142*, 2025.
- [6] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafford. Think you have solved question answering? try ARC, the AI2 reasoning challenge, 2018. URL <https://arxiv.org/abs/1803.05457>.
- [7] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL <https://arxiv.org/abs/2110.14168>.
- [8] John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12(7), 2011.
- [9] Yuchao Gu, Xintao Wang, Jay Zhangjie Wu, Yujun Shi, Yunpeng Chen, Zihan Fan, Wuyou Xiao, Rui Zhao, Shuning Chang, Weijia Wu, et al. Mix-of-show: Decentralized low-rank adaptation for multi-concept customization of diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 15890–15902, 2023.
- [10] Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned stochastic tensor optimization. In *International Conference on Machine Learning (ICML)*, pages 1842–1850. PMLR, 2018.
- [11] Soufiane Hayou, Nikhil Ghosh, and Bin Yu. Lora+: Efficient low rank adaptation of large models. In *International Conference on Machine Learning (ICML)*, pages 17783–17806. PMLR, 2024.
- [12] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP (1)*, 2021.
- [13] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in neural information processing systems (NeurIPS)*, volume 30, 2017.
- [14] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, page 3, 2022.
- [15] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mistral 7b, 2023. URL <https://arxiv.org/abs/2310.06825>.

- [16] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [17] Frederik Kunstner, Philipp Hennig, and Lukas Balles. Limitations of the empirical fisher approximation for natural gradient descent. *Advances in neural information processing systems*, 32, 2019.
- [18] Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- [19] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *International conference on machine learning*, pages 2408–2417. PMLR, 2015.
- [20] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *International conference on machine learning (ICML)*, pages 2408–2417. PMLR, 2015.
- [21] Zhanfeng Mo, Long-Kai Huang, and Sinno Jialin Pan. Parameter and memory efficient pretraining via low-rank riemannian optimization. In *International Conference on Learning Representations (ICLR)*, 2025.
- [22] Depen Morwani, Itai Shapira, Nikhil Vyas, Sham M Kakade, Lucas Janson, et al. A new perspective on shampoo’s preconditioner. In *International Conference on Learning Representations (ICLR)*, 2024.
- [23] Linyong Nan, Dragomir Radev, Rui Zhang, Amrit Rau, Abhinand Sivaprasad, Chiachun Hsieh, Xiangru Tang, Aadit Vyas, Neha Verma, Pranav Krishna, et al. Dart: Open-domain structured data record to text generation. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 432–447, 2021.
- [24] Jekaterina Novikova, Ondřej Dušek, and Verena Rieser. The e2e dataset: New challenges for end-to-end generation. *arXiv preprint arXiv:1706.09254*, 2017.
- [25] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems (NeurIPS)*, volume 32, 2019.
- [26] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [27] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning (ICML)*, pages 8748–8763. PMLR, 2021.
- [28] Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *International Conference on Machine Learning (ICML)*, pages 4596–4604. PMLR, 2018.
- [29] Stephen Tu, Ross Boczar, Max Simchowitz, Mahdi Soltanolkotabi, and Ben Recht. Low-rank solutions of linear matrix equations via procrustes flow. In *International conference on machine learning*, pages 964–973. PMLR, 2016.
- [30] Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham M Kakade. Soap: Improving and stabilizing shampoo using adam for language modeling. In *International Conference on Learning Representations (ICLR)*, 2025.
- [31] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. Glue: A multi-task benchmark and analysis platform for natural language understanding, 2019. URL <https://arxiv.org/abs/1804.07461>.

- [32] Shaowen Wang, Linxi Yu, and Jian Li. Lora-ga: Low-rank adaptation with gradient approximation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 54905–54931, 2024.
- [33] Zhengbo Wang, Jian Liang, Ran He, Zilei Wang, and Tieniu Tan. Lora-pro: Are low-rank adapters properly optimized? In *International Conference on Learning Representations (ICLR)*, 2025.
- [34] Ke Wei, Jian-Feng Cai, Tony F Chan, and Shingyu Leung. Guarantees of riemannian optimization for low rank matrix recovery. *SIAM Journal on Matrix Analysis and Applications*, 37(3): 1198–1222, 2016.
- [35] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [36] Jui-Nan Yen, Si Si, Zhao Meng, Felix Yu, Sai Surya Duvvuri, Inderjit S Dhillon, Cho-Jui Hsieh, and Sanjiv Kumar. Lora done rite: Robust invariant transformation equilibration for lora optimization. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [37] Fangzhao Zhang and Mert Pilanci. Riemannian preconditioned lora for fine-tuning foundation models. In *International Conference on Machine Learning (ICML)*, 2024.
- [38] Yuanhe Zhang, Fanghui Liu, and Yudong Chen. Lora-one: One-step full gradient could suffice for fine-tuning large language models, provably and efficiently. In *International Conference on Machine Learning (ICML)*, 2025.
- [39] Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuandong Tian. Galore: Memory-efficient llm training by gradient low-rank projection. In *International Conference on Machine Learning (ICML)*, pages 61121–61143. PMLR, 2024.
- [40] Zhenyu Zhu, Yongtao Wu, Quanquan Gu, and Volkan Cevher. Imbalance-regularized lora: A plug-and-play method for improving fine-tuning of foundation models. In *Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning*, 2024.

Contents

A	Notation	14
B	Proof of Theoretical Results	14
B.1	Computation of Jacobian	14
B.2	Proof of Proposition B.2 (Kernel of $J_{\mathcal{G}}$)	16
B.3	Optimality of $\mathcal{H}_t$ -Projection (Solution 1)	17
B.4	Orthogonal Projection to Tangent Space	17
B.5	Proof of Theorem 3.2 (closed-form factor update)	19
C	Algorithm	20
D	Computational and Memory Complexity Analysis of SoLoRA	21
E	Supplementary Experiments of GPT-2 Fine-tuning	22
E.1	Parameter Settings	22
E.2	Cross-rank ablation on GPT-2 small	22
E.3	Training Efficiency Comparison	23
F	Supplementary Experiments of Diffusion Model Fine-tuning	23

A Notation

Table 7 summarizes the notation used throughout the paper. Calligraphic letters denote linear operators on $\mathbb{R}^{m \times n}$ (e.g., $\mathcal{F}_t, \mathcal{H}_t$), while bold letters denote matrices (e.g., $\mathbf{B}_t, \mathbf{A}_t, \mathbf{L}_t, \mathbf{R}_t, \mathbf{G}_t$).

B Proof of Theoretical Results

B.1 Computation of Jacobian

Proposition B.1 (Computation of $J_{\mathcal{G}}$ and $J_{\mathcal{G}}^*$). *Let $[\mathbf{B}, \mathbf{A}]$ be a pair of low-rank factors with $\mathbf{B} \in \mathbb{R}^{m \times r}, \mathbf{A} \in \mathbb{R}^{r \times n}$. Define the generator $\mathcal{G} : [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}] \rightarrow \mathbb{R}^{m \times n}$ by $\mathcal{G}([\mathbf{B}, \mathbf{A}]) = \mathbf{B}\mathbf{A}$. Denote the Jacobian of $\mathcal{G}$ by $J_{\mathcal{G}}$ and its adjoint by $J_{\mathcal{G}}^*$. Then, for any $[\mathbf{P}, \mathbf{Q}] \in [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$ and any $\mathbf{C} \in \mathbb{R}^{m \times n}$,*

- $J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])[\mathbf{P}, \mathbf{Q}] = \mathbf{P}\mathbf{A} + \mathbf{B}\mathbf{Q}$,
- $J_{\mathcal{G}}^*([\mathbf{B}, \mathbf{A}])(\mathbf{C}) = [\mathbf{C}\mathbf{A}^\top, \mathbf{B}^\top\mathbf{C}]$,
- $J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])J_{\mathcal{G}}^*([\mathbf{B}, \mathbf{A}])(\mathbf{C}) = \mathbf{C}\mathbf{A}^\top\mathbf{A} + \mathbf{B}\mathbf{B}^\top\mathbf{C}$.
- $J_{\mathcal{G}}^*([\mathbf{B}, \mathbf{A}])J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])[\mathbf{P}, \mathbf{Q}] = [\mathbf{P}\mathbf{A}\mathbf{A}^\top + \mathbf{B}\mathbf{Q}\mathbf{A}^\top, \mathbf{B}^\top\mathbf{P}\mathbf{A} + \mathbf{B}^\top\mathbf{B}\mathbf{Q}]$.

Proof. The Jacobian operator $J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])[\mathbf{P}, \mathbf{Q}] : [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}] \rightarrow \mathbb{R}^{m \times n}$ represents the derivative of $\mathcal{G}$ at $[\mathbf{B}, \mathbf{A}]$ along the direction $[\mathbf{P}, \mathbf{Q}]$. Similarly, $J_{\mathcal{G}}^*([\mathbf{B}, \mathbf{A}])(\mathbf{C}) : \mathbb{R}^{m \times n} \rightarrow [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$ is the adjoint of $J_{\mathcal{G}}$ at $[\mathbf{B}, \mathbf{A}]$ along the direction $\mathbf{C}$. For more details, see [1, § 6.1].

- (i) The computation of $J_{\mathcal{G}}$. Let $\mathbf{B}(t) : \mathbb{R} \rightarrow \mathbb{R}^{m \times r}$ and $\mathbf{A}(t) : \mathbb{R} \rightarrow \mathbb{R}^{r \times n}$ be differentiable curves with $\mathbf{B}(0) = \mathbf{B}$ and $\mathbf{A}(0) = \mathbf{A}$. By the chain rule, the Jacobian of $\mathcal{G}$ at $[\mathbf{B}, \mathbf{A}]$ along these curves is

$$\begin{aligned}
 J_{\mathcal{G}}([\mathbf{B}(t), \mathbf{A}(t)])(\dot{\mathbf{B}}(t), \dot{\mathbf{A}}(t)) \Big|_{t=0} &= \left[\frac{d\mathcal{G}([\mathbf{B}, \mathbf{A}])}{d\mathbf{B}} \right] \dot{\mathbf{B}}(t) \Big|_{t=0} + \left[\frac{d\mathcal{G}([\mathbf{B}, \mathbf{A}])}{d\mathbf{A}} \right] \dot{\mathbf{A}}(t) \Big|_{t=0} \\
 &= \dot{\mathbf{B}}(t)\mathbf{A}(t) \Big|_{t=0} + \mathbf{B}(t)\dot{\mathbf{A}}(t) \Big|_{t=0} \\
 &= \dot{\mathbf{B}}(0)\mathbf{A} + \mathbf{B}\dot{\mathbf{A}}(0),
 \end{aligned}$$

Table 7: Notation used throughout the paper.

Symbol	Meaning
<i>Dimensions and indices</i>	
m, n	Output and input dimensions of the adapted weight $\mathbf{W} \in \mathbb{R}^{m \times n}$
r	LoRA rank, $r \ll \min(m, n)$
t	Optimization step index
<i>Matrices (in bold)</i>	
$\mathbf{W}_0, \mathbf{W}_t$	Frozen pretrained weight; LoRA increment at step t , $\mathbf{W}_t = \mathbf{B}_t \mathbf{A}_t$
$\mathbf{B}_t, \mathbf{A}_t$	LoRA factors, $\mathbf{B}_t \in \mathbb{R}^{m \times r}$, $\mathbf{A}_t \in \mathbb{R}^{r \times n}$
$\mathbf{G}_t$	Stochastic gradient $\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_0 + \mathbf{W}_t)$ on $\mathbf{W}$
$\mathbf{G}_{\mathbf{B}_t}, \mathbf{G}_{\mathbf{A}_t}$	Factor gradients $\mathbf{G}_t \mathbf{A}_t^\top$ and $\mathbf{B}_t^\top \mathbf{G}_t$
$\Delta_{\mathbf{B}_t}, \Delta_{\mathbf{A}_t}$	Factor updates: $\mathbf{B}_{t+1} = \mathbf{B}_t - \eta_t \Delta_{\mathbf{B}_t}$, similarly for $\mathbf{A}$
Δ_t	Induced $\mathbf{W}$ -update $\Delta_{\mathbf{B}_t} \mathbf{A}_t + \mathbf{B}_t \Delta_{\mathbf{A}_t}$
$\mathbf{l}_t, \mathbf{r}_t$	Adafactor row/column statistics, $\mathbf{l}_t \in \mathbb{R}^m$, $\mathbf{r}_t \in \mathbb{R}^n$
$\mathbf{L}_t, \mathbf{R}_t$	Adafactor diagonal preconditioner factors (Eq. (3.1))
$\mathbf{L}_{\text{Sh},t}, \mathbf{R}_{\text{Sh},t}$	Shampoo full-covariance factors $\sum_s \mathbf{G}_s \mathbf{G}_s^\top$, $\sum_s \mathbf{G}_s^\top \mathbf{G}_s$
$\mathbf{X}_t$	Free $r \times r$ matrix parameterizing the $\ker(J_{\mathcal{G}})$ orbit (Lemma B.1)
$\tilde{\mathbf{P}}_{\mathbf{B}_t}, \tilde{\mathbf{Q}}_{\mathbf{A}_t}$	Auxiliary $\mathbf{L}_t^{1/2}/\mathbf{R}_t^{1/2}$ -weighted projector matrices in Theorem 3.2
<i>Operators (in calligraphic)</i>	
$\mathcal{F}_t$	Empirical Fisher operator on $\mathbf{W}$, $\mathcal{F}_t : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$
$\mathcal{H}_t$	Operator square root, $\mathcal{H}_t = \mathcal{F}_t^{1/2}$; on Adafactor: $\mathcal{H}_t \mathbf{Y} = \mathbf{L}_t^{1/2} \mathbf{Y} \mathbf{R}_t^{1/2}$
$\mathcal{E}_t$	Empirical Fisher on the factor space $[\mathbf{B}_t, \mathbf{A}_t]$, related to $\mathcal{F}_t$ via $\mathcal{E}_t = J_{\mathcal{G}}^* \mathcal{F}_t J_{\mathcal{G}}$ (Eq. (4))
$\tilde{\mathcal{P}}_{\mathbb{T}_t}, \tilde{\mathcal{P}}_{\mathbb{T}_t}^\perp$	$\mathcal{H}_t$ -orthogonal projector onto $\mathbb{T}_t$ and its complement
$\mathcal{P}_{\mathbb{T}_t}$	Frobenius (Euclidean) orthogonal projector onto $\mathbb{T}_t$
$J_{\mathcal{G}}, J_{\mathcal{G}}^*, J_{\mathcal{G}}^\dagger$	Jacobian of $\mathcal{G} : [\mathbf{B}, \mathbf{A}] \mapsto \mathbf{B}\mathbf{A}$, its adjoint, and Moore–Penrose pseudoinverse
<i>Subspaces and inner products</i>	
$\mathbb{T}_t$	$\text{Im}(J_{\mathcal{G}}) = \{\mathbf{P}\mathbf{A}_t + \mathbf{B}_t\mathbf{Q}\} \subset \mathbb{R}^{m \times n}$, range of one LoRA step
$\mathcal{M}_r$	Manifold of rank- r matrices in $\mathbb{R}^{m \times n}$
$\langle \cdot, \cdot \rangle$	Frobenius inner product on $\mathbb{R}^{m \times n}$
$\langle \cdot, \cdot \rangle_{\mathcal{H}_t}, \ \cdot\ _{\mathcal{H}_t}$	$\mathcal{H}_t$ -weighted inner product and norm (Eq. (9))
$\tilde{\mathbf{G}}_t$	Adafactor-preconditioned direction $\mathcal{H}_t^{-1} \mathbf{G}_t = \mathbf{L}_t^{-1/2} \mathbf{G}_t \mathbf{R}_t^{-1/2}$
<i>Other</i>	
η_t	Learning rate at step t
β_1, β_2	EMA decay rates for Adafactor statistics $\mathbf{l}_t, \mathbf{r}_t$
$\odot$	Hadamard (elementwise) product
$\text{diag}(\mathbf{v})$	Diagonal matrix with $\mathbf{v}$ on the diagonal
$\mathcal{L}$	Loss function

where $\dot{\mathbf{B}}(t)$ and $\dot{\mathbf{A}}(t)$ denote the derivatives of $\mathbf{B}(t)$ and $\mathbf{A}(t)$ with respect to t . The second line follows because $\mathcal{G}([\mathbf{B}, \mathbf{A}]) = \mathbf{B}\mathbf{A}$, hence $\frac{d\mathcal{G}([\mathbf{B}, \mathbf{A}])}{d\mathbf{B}}$ and $\frac{d\mathcal{G}([\mathbf{B}, \mathbf{A}])}{d\mathbf{A}}$ are both linear operators.

Since $\dot{\mathbf{B}}(0)$ and $\dot{\mathbf{A}}(0)$ are arbitrary, for any $[\mathbf{P}, \mathbf{Q}] \in [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$, we obtain

$$J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])[\mathbf{P}, \mathbf{Q}] = \mathbf{P}\mathbf{A} + \mathbf{B}\mathbf{Q}.$$

- (ii) The computation of $J_{\mathcal{G}}^*$. For brevity, write $J_{\mathcal{G}}[\mathbf{P}, \mathbf{Q}]$ for $J_{\mathcal{G}}([\mathbf{B}, \mathbf{A}])[\mathbf{P}, \mathbf{Q}]$ and $J_{\mathcal{G}}^*(\mathbf{C})$ for $J_{\mathcal{G}}^*([\mathbf{B}, \mathbf{A}])(\mathbf{C})$. By definition of the adjoint (with respect to the Frobenius inner product), for any $[\mathbf{P}, \mathbf{Q}] \in (\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n})$ and $\mathbf{C} \in \mathbb{R}^{m \times n}$,

$$\langle J_{\mathcal{G}}[\mathbf{P}, \mathbf{Q}], \mathbf{C} \rangle = \langle [\mathbf{P}, \mathbf{Q}], J_{\mathcal{G}}^*(\mathbf{C}) \rangle.$$

For the left-hand side,

$$\begin{aligned} \langle J_{\mathcal{G}}[\mathbf{P}, \mathbf{Q}], \mathbf{C} \rangle &= \langle \mathbf{P}\mathbf{A} + \mathbf{B}\mathbf{Q}, \mathbf{C} \rangle \\ &= \langle \mathbf{P}\mathbf{A}, \mathbf{C} \rangle + \langle \mathbf{B}\mathbf{Q}, \mathbf{C} \rangle \\ &= \langle \mathbf{P}, \mathbf{C}\mathbf{A}^\top \rangle + \langle \mathbf{Q}, \mathbf{B}^\top \mathbf{C} \rangle. \end{aligned}$$

For the right-hand side, writing $J_G^*(C) = [C_1, C_2]$, then

$$\begin{aligned}\langle [P, Q], J_G^*(C) \rangle &= \langle [P, Q], [C_1, C_2] \rangle \\ &= \langle P, C_1 \rangle + \langle Q, C_2 \rangle.\end{aligned}$$

Hence $C_1 = CA^\top$ and $C_2 = B^\top C$, and therefore $J_G^*([B, A])(C) = [CA^\top, B^\top C]$.

(iii) By (i)–(ii), $J_G([B, A])J_G^*([B, A])(C) = J_G([B, A])[CA^\top, B^\top C] = CA^\top A + BB^\top C$ as claimed.

(iv) By (i)–(ii), $J_G^*([B, A])J_G([B, A])[P, Q] = J_G^*([B, A])(PA + BQ) = [(PA + BQ)A^\top, B^\top(PA + BQ)] = [PAA^\top + BQA^\top, B^\top PA + B^\top BQ]$ as claimed.

□

B.2 Proof of Proposition B.2 (Kernel of J_G)

Proposition B.2 (Kernel of J_G from factorization redundancy). *For $B_t \in \mathbb{R}^{m \times r}$ of column rank r and $A_t \in \mathbb{R}^{r \times n}$ of row rank r , the kernel of the Jacobian operator $J_G = J_G([B_t, A_t])$ is*

$$\ker(J_G) = \{ [B_t X, -X A_t] : X \in \mathbb{R}^{r \times r} \},$$

which is an r^2 -dimensional linear subspace of $[\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$. Consequently, $\text{rank}(J_G) = (m + n)r - r^2$.

Proof. We prove the three statements in turn: (i) the indicated set is contained in $\ker(J_G)$; (ii) every kernel element has this form; (iii) the dimension equals r^2 . The rank statement then follows by the rank–nullity theorem applied to $J_G : [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}] \rightarrow \mathbb{R}^{m \times n}$, whose domain has dimension $(m + n)r$.

(i) **Inclusion.** For any $X \in \mathbb{R}^{r \times r}$, by the definition of J_G in (1),

$$J_G[B_t X, -X A_t] = (B_t X)A_t + B_t(-X A_t) = B_t X A_t - B_t X A_t = 0.$$

Hence $[B_t X, -X A_t] \in \ker(J_G)$ for every X .

(ii) **Reverse inclusion.** Suppose $[P, Q] \in \ker(J_G)$, i.e., $PA_t + B_t Q = 0$, equivalently

$$PA_t = -B_t Q. \tag{15}$$

Since B_t has column rank r and A_t has row rank r , define their one-sided pseudoinverses

$$B_t^+ := (B_t^\top B_t)^{-1} B_t^\top \in \mathbb{R}^{r \times m}, \quad A_t^+ := A_t^\top (A_t A_t^\top)^{-1} \in \mathbb{R}^{n \times r},$$

satisfying $B_t^+ B_t = I_r$ and $A_t A_t^+ = I_r$. Set $X := B_t^+ P \in \mathbb{R}^{r \times r}$.

Left-multiplying (15) by B_t^+ gives $X A_t = -Q$, i.e.,

$$Q = -X A_t.$$

Substituting back into (15) and right-multiplying by A_t^+ gives

$$P = PA_t A_t^+ = -B_t Q A_t^+ = B_t X A_t A_t^+ = B_t X.$$

Therefore $[P, Q] = [B_t X, -X A_t]$, as claimed.

(iii) **Dimension.** The map $\Phi : \mathbb{R}^{r \times r} \rightarrow [\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]$ defined by $\Phi(X) = [B_t X, -X A_t]$ is linear. It is injective: if $\Phi(X) = 0$ then $B_t X = 0$, and the column-rank condition on B_t gives $X = 0$. Hence $\dim(\text{im } \Phi) = \dim(\mathbb{R}^{r \times r}) = r^2$. By (i)–(ii), $\text{im } \Phi = \ker(J_G)$, so $\dim \ker(J_G) = r^2$.

Rank. By the rank–nullity theorem,

$$\text{rank}(J_G) = \dim([\mathbb{R}^{m \times r}, \mathbb{R}^{r \times n}]) - \dim \ker(J_G) = (m + n)r - r^2.$$

Equivalently, $\text{rank}(J_G) = \dim \mathcal{M}_r$, the dimension of the rank- r manifold at $W_t = B_t A_t$, consistent with $\text{im}(J_G) = \mathbb{T}_t$ (Proposition B.6). □

B.3 Optimality of $\mathcal{H}_t$ -Projection (Solution 1)

Proposition B.3 (Common $\mathbf{W}$ -update across the affine solution set). *Let $\mathcal{H}_t : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ be a symmetric positive-definite operator, inducing the inner product $\langle \mathbf{X}, \mathbf{Y} \rangle_{\mathcal{H}_t} := \langle \mathcal{H}_t \mathbf{X}, \mathbf{Y} \rangle$ on $\mathbb{R}^{m \times n}$. Let $\mathbb{T}_t := \text{range}(J_{\mathcal{G}}) \subset \mathbb{R}^{m \times n}$ and $\tilde{\mathbf{G}}_t := \mathcal{H}_t^{-1} \mathbf{G}_t$. Then every $[\Delta_{B_t}, \Delta_{A_t}]$ in the affine solution set of (10) satisfies*

$$J_{\mathcal{G}}[\Delta_{B_t}, \Delta_{A_t}] = \tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{\mathbf{G}}_t),$$

where $\tilde{\mathcal{P}}_{\mathbb{T}_t}$ is the $\mathcal{H}_t$ -orthogonal projector onto $\mathbb{T}_t$, characterized by $\langle \tilde{\mathcal{P}}_{\mathbb{T}_t}(\mathbf{X}) - \mathbf{X}, \mathbf{Y} \rangle_{\mathcal{H}_t} = 0$ for all $\mathbf{X} \in \mathbb{R}^{m \times n}, \mathbf{Y} \in \mathbb{T}_t$.

Proof. Equip $V := \mathbb{R}^{m \times n}$ with the inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}_t}$. Since $\mathcal{H}_t$ is SPD, this is a genuine inner product, and V is a finite-dimensional Hilbert space. The subspace $\mathbb{T}_t$ is finite-dimensional, hence closed in V , and admits a unique $\mathcal{H}_t$ -orthogonal decomposition $V = \mathbb{T}_t \oplus \mathbb{T}_t^{\perp}$, defining $\tilde{\mathcal{P}}_{\mathbb{T}_t}$.

Let $[\Delta_{B_t}, \Delta_{A_t}]$ be any solution of (10). Using $J_{\mathcal{G}}^*(\mathbf{G}_t) = J_{\mathcal{G}}^* \mathcal{H}_t \tilde{\mathbf{G}}_t$, the equation rewrites as $J_{\mathcal{G}}^* \mathcal{H}_t (J_{\mathcal{G}}[\Delta_{B_t}, \Delta_{A_t}] - \tilde{\mathbf{G}}_t) = 0$, i.e. $J_{\mathcal{G}}[\Delta_{B_t}, \Delta_{A_t}] - \tilde{\mathbf{G}}_t \in \mathbb{T}_t^{\perp}$ (since $\ker(J_{\mathcal{G}}^* \mathcal{H}_t) = \mathbb{T}_t^{\perp}$ by adjointness). Decomposing $\tilde{\mathbf{G}}_t = \tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{\mathbf{G}}_t) + \tilde{\mathcal{P}}_{\mathbb{T}_t}^{\perp}(\tilde{\mathbf{G}}_t)$, this gives $J_{\mathcal{G}}[\Delta_{B_t}, \Delta_{A_t}] - \tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{\mathbf{G}}_t) \in \mathbb{T}_t^{\perp}$. But the left-hand side also lies in $\mathbb{T}_t$ (both summands do), and $\mathbb{T}_t \cap \mathbb{T}_t^{\perp} = \{0\}$, so $J_{\mathcal{G}}[\Delta_{B_t}, \Delta_{A_t}] = \tilde{\mathcal{P}}_{\mathbb{T}_t}(\tilde{\mathbf{G}}_t)$. $\square$

Comparison with LoRA-Pro. When $\mathcal{H}_t = I$, the $\mathcal{H}_t$ -orthogonal projector $\tilde{\mathcal{P}}_{\mathbb{T}_t}$ collapses to the Frobenius projector $\mathcal{P}_{\mathbb{T}_t}$, and the common $\mathbf{W}$ -update reduces to $\mathcal{P}_{\mathbb{T}_t}(\mathbf{G}_t)$ – the LoRA-Pro update. For non-trivial $\mathcal{H}_t$, the two projections operate in different geometries (residuals lie in $\mathbb{T}_t^{\perp}$ vs. $\mathbb{T}_t^{\perp}$).

B.4 Orthogonal Projection to Tangent Space

Relation to the main-text subspace $\mathbb{T}_t$. In the main text, $\mathbb{T}_t = \text{Im}(J_{\mathcal{G}})$ is treated as a linear subspace of $\mathbb{R}^{m \times n}$ (§ 2.2), with no manifold structure assumed. Whenever $\mathbf{W}_t = \mathbf{B}_t \mathbf{A}_t$ has rank exactly r (i.e., $\mathbf{W}_t \in \mathcal{M}_r$), $\text{Im}(J_{\mathcal{G}})$ coincides with the tangent space $\mathbb{T}_{\mathbf{W}_t}$ of the rank- r manifold $\mathcal{M}_r$ at $\mathbf{W}_t$ (Proposition B.6). The propositions in this subsection adopt the manifold viewpoint $\mathbb{T}_{\mathbf{W}}$ since the SVD-based proofs are most natural under it; the resulting projection formulas apply directly to the main-text subspace $\mathbb{T}_t$.

In this subsection, we derive the orthogonal projection onto the tangent space under both the standard metric and the weighted metric. The specific forms of $\mathbf{L}_t$ and $\mathbf{R}_t$ are presented here and will not be repeated in subsequent propositions and proofs. For the sake of simplicity, the subscript t will be omitted in this subsection.

$$\begin{aligned} \mathbf{L}_t &= \text{diag}(\mathbf{l}_t / \sqrt{\|\mathbf{l}_t\|_1}) \quad \text{with} \quad \mathbf{l}_t = \beta_2 \mathbf{l}_{t-1} + (1 - \beta_2) \sum_{j=1}^n (\mathbf{G}_t \odot \mathbf{G}_t)_{i,j}, \\ \mathbf{R}_t &= \text{diag}(\mathbf{r}_t / \sqrt{\|\mathbf{r}_t\|_1}) \quad \text{with} \quad \mathbf{r}_t = \beta_3 \mathbf{r}_{t-1} + (1 - \beta_3) \sum_{i=1}^m (\mathbf{G}_t \odot \mathbf{G}_t)_{i,j}, \end{aligned} \tag{16}$$

where $\odot$ denotes the Hadamard (elementwise) product and $\mathbf{G}_t = \nabla \mathcal{L}(\mathbf{W}_0 + \mathbf{W}_t)$.

Proposition B.4 (Orthogonal Projection to Tangent Space Under the Standard Metric). *Let $\mathbf{W} \in \mathcal{M}_r$ be a rank- r matrix with a low-rank decomposition $\mathbf{W} = \mathbf{B} \mathbf{A}$, where $\mathbf{B} \in \mathbb{R}^{m \times r}, \mathbf{A} \in \mathbb{R}^{r \times n}$. Denote by $\mathbb{T}_{\mathbf{W}}$ the tangent space of the smooth manifold $\mathcal{M}_r$ at the point $\mathbf{W}$. Then, the orthogonal projection of any matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$ onto $\mathbb{T}_{\mathbf{W}}$ is given by*

$$\mathcal{P}_{\mathbb{T}_{\mathbf{W}}}(\mathbf{Z}) = \mathbf{B}(\mathbf{B}^{\top} \mathbf{B})^{-1} \mathbf{B}^{\top} \mathbf{Z} + \mathbf{Z} \mathbf{A}^{\top} (\mathbf{A} \mathbf{A}^{\top})^{-1} \mathbf{A} - \mathbf{B}(\mathbf{B}^{\top} \mathbf{B})^{-1} \mathbf{B}^{\top} \mathbf{Z} \mathbf{A}^{\top} (\mathbf{A} \mathbf{A}^{\top})^{-1} \mathbf{A}.$$

Proof. Suppose $\mathbf{W}$ has a compact singular value decomposition, given by $\mathbf{W} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^{\top}$, where $\mathbf{U} \in \mathbb{R}^{m \times r}, \mathbf{\Sigma} \in \mathbb{R}^{r \times r}, \mathbf{V} \in \mathbb{R}^{n \times r}$. Then the tangent space $\mathbb{T}_{\mathbf{W}}$ at $\mathbf{W}$ is characterized as

$$\mathbb{T}_{\mathbf{W}} = \{\mathbf{U} \mathbf{M}^{\top} + \mathbf{N} \mathbf{V}^{\top}, \text{ for } \mathbf{M} \in \mathbb{R}^{m \times r}, \mathbf{N} \in \mathbb{R}^{n \times r}\}.$$

Therefore, the orthogonal projection of $\mathbf{Z}$ onto $\mathbb{T}_{\mathbf{W}}$ is known to be [34]

$$\mathcal{P}_{\mathbb{T}_{\mathbf{W}}}(\mathbf{Z}) = \mathbf{U}\mathbf{U}^\top \mathbf{Z} + \mathbf{Z}\mathbf{V}^\top \mathbf{V} - \mathbf{U}\mathbf{U}^\top \mathbf{Z}\mathbf{V}^\top \mathbf{V}. \quad (17)$$

Since the columns of $\mathbf{B}$ and $\mathbf{U}$ span the same column space (i.e., the column space of $\mathbf{W}$), then there exists an invertible matrix $\mathbf{S} \in \mathbb{R}^{r \times r}$ such that $\mathbf{B} = \mathbf{U}\mathbf{S}$ and $\mathbf{U} = \mathbf{B}\mathbf{S}^{-1}$. Using this relation, we have

$$\mathbf{U}^\top \mathbf{U} = (\mathbf{B}\mathbf{S}^{-1})^\top \mathbf{B}\mathbf{S}^{-1} = \mathbf{S}^{-\top} (\mathbf{B}^\top \mathbf{B}) \mathbf{S}^{-1}.$$

Since $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r$, it follows that

$$\mathbf{S}^{-\top} (\mathbf{B}^\top \mathbf{B}) \mathbf{S}^{-1} = \mathbf{I}_r \implies \mathbf{B}^\top \mathbf{B} = \mathbf{S}^\top \mathbf{S}.$$

Using this, we compute $\mathbf{U}\mathbf{U}^\top$

$$\mathbf{U}\mathbf{U}^\top = \mathbf{B}\mathbf{S}^{-1}\mathbf{S}^{-\top} \mathbf{B} = \mathbf{B}(\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{B}^\top = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \quad (18)$$

Similarly, since the rows of $\mathbf{A}$ and the columns of $\mathbf{V}$ span the same row space (i.e., the row space of $\mathbf{W}$), there exists an invertible matrix $\mathbf{Q} \in \mathbb{R}^{r \times r}$ such that $\mathbf{A} = \mathbf{Q}\mathbf{V}^\top$ and $\mathbf{V}^\top = \mathbf{Q}^{-1}\mathbf{A}$. Further, using $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$, we obtain

$$\mathbf{V}^\top \mathbf{V} = \mathbf{Q}^{-1}(\mathbf{A}\mathbf{A}^\top)\mathbf{Q}^{-\top} = \mathbf{I}_r,$$

hence $\mathbf{A}\mathbf{A}^\top = \mathbf{Q}\mathbf{Q}^\top$ and

$$\mathbf{V}\mathbf{V}^\top = \mathbf{A}^\top \mathbf{Q}^{-\top} \mathbf{Q}^{-1} \mathbf{A} = \mathbf{A}^\top (\mathbf{Q}\mathbf{Q}^\top)^{-1} \mathbf{A} = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{A} \quad (19)$$

Substituting (18) and (19) into (17) yields

$$\mathcal{P}_{\mathbb{T}_{\mathbf{W}}}(\mathbf{Z}) = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{Z} + \mathbf{Z}\mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{A} - \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{Z}\mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} \mathbf{A}.$$

□

Proposition B.5 (Orthogonal Projection onto the Tangent Space Under the Weighted Metric). *Let $\mathbf{W} \in \mathcal{M}_r$ has a low-rank decomposition $\mathbf{W} = \mathbf{B}\mathbf{A}$, where $\mathbf{B} \in \mathbb{R}^{m \times r}$, $\mathbf{A} \in \mathbb{R}^{r \times n}$. Denote the tangent space of the Riemannian manifold $\mathcal{M}_r$ at the point $\mathbf{W}$ as $\mathbb{T}_{\mathbf{W}}$. The weighted metric is defined as $\langle \mathbf{Y}, \mathbf{Z} \rangle_{\mathbf{H}} = \langle \mathbf{L}^{\frac{1}{2}} \mathbf{Y} \mathbf{R}^{\frac{1}{2}}, \mathbf{Z} \rangle$ for any $\mathbf{Y}, \mathbf{Z} \in \mathbb{R}^{m \times n}$. Then, the orthogonal projection of any matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$ onto $\mathbb{T}_{\mathbf{W}}$ under the weighed metric is given by*

$$\begin{aligned} \mathcal{P}_{\mathbb{T}_{\mathbf{W}}}(\mathbf{Z}) &= \mathbf{B}(\mathbf{B}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{Z} + \mathbf{Z} \mathbf{R}^{\frac{1}{2}} \mathbf{A}^\top (\mathbf{A} \mathbf{R}^{\frac{1}{2}} \mathbf{A}^\top)^{-1} \mathbf{A} \\ &\quad - \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{Z} \mathbf{R}^{\frac{1}{2}} \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1} \mathbf{A}. \end{aligned}$$

Proof. This proof is inspired by [3]. Here, we briefly provide a sketch of the proof.

- (i) *The new orthonormal basis under the weighted metric.* Let $\mathbf{W} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be a compact SVD with $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r] \in \mathbb{R}^{m \times r}$, $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r] \in \mathbb{R}^{n \times r}$. Normalize the singular vectors under the weighted vector

$$\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{L}^{\frac{1}{2}}} = \langle \mathbf{L}^{\frac{1}{2}} \mathbf{x}, \mathbf{y} \rangle \text{ in } \mathbb{R}^m \quad \text{and} \quad \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{R}^{\frac{1}{2}}} = \langle \mathbf{R}^{\frac{1}{2}} \mathbf{x}, \mathbf{y} \rangle \text{ in } \mathbb{R}^n$$

to obtain

$$\begin{aligned} \tilde{\mathbf{U}} &= \mathbf{U}(\mathbf{U}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{U})^{-\frac{1}{2}} := [\tilde{\mathbf{u}}_1, \tilde{\mathbf{u}}_2, \dots, \tilde{\mathbf{u}}_r] \in \mathbb{R}^{m \times r}, \\ \tilde{\mathbf{V}} &= \mathbf{V}(\mathbf{V}^\top \mathbf{R}^{\frac{1}{2}} \mathbf{V})^{-\frac{1}{2}} := [\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \dots, \tilde{\mathbf{v}}_r] \in \mathbb{R}^{n \times r}. \end{aligned}$$

Next, we extend $\tilde{\mathbf{U}}$ and $\tilde{\mathbf{V}}$ to full orthonormal basis of $(\mathbb{R}^m, \langle \cdot, \cdot \rangle_{\mathbf{L}^{\frac{1}{2}}})$ and $(\mathbb{R}^n, \langle \cdot, \cdot \rangle_{\mathbf{R}^{\frac{1}{2}}})$ respectively. Then, an orthonormal basis of $\mathbb{T}_{\mathbf{W}}$ with respect to $\langle \cdot, \cdot \rangle_{\mathbf{H}_t}$ is $\{\tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top\}_{\min\{i,j\} \leq r}$.

- (ii) *Orthogonal projection represented by the new orthonormal basis.* Using the orthonormal bases $\tilde{\mathbf{U}}$ and $\tilde{\mathbf{V}}$, the projection of $\mathbf{Z}$ onto $\mathbb{T}_{\mathbf{W}}$ is expressed as:

$$\begin{aligned} \tilde{\mathcal{P}}_{\mathbb{T}_{\mathbf{W}}}(\mathbf{Z}) &= \sum_{(i,j): \min\{i,j\} \leq r} \langle \mathbf{Z}, \tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top \rangle_{\mathbf{H}_t} \cdot \tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top = \sum_{(i,j): \min\{i,j\} \leq r} \langle \mathbf{L}^{\frac{1}{2}} \mathbf{Z} \mathbf{R}^{\frac{1}{2}}, \tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top \rangle \cdot \tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top \\ &= \sum_{(i,j): \min\{i,j\} \leq r} \tilde{\mathbf{u}}_i^\top \mathbf{L}^{\frac{1}{2}} \mathbf{Z} \mathbf{R}^{\frac{1}{2}} \tilde{\mathbf{v}}_j \cdot \tilde{\mathbf{u}}_i \tilde{\mathbf{v}}_j^\top \\ &= \tilde{\mathbf{U}} \tilde{\mathbf{U}}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{Z} + \mathbf{Z} \mathbf{R}^{\frac{1}{2}} \tilde{\mathbf{V}} \tilde{\mathbf{V}}^\top - \tilde{\mathbf{U}} \tilde{\mathbf{U}}^\top \mathbf{L}^{\frac{1}{2}} \mathbf{Z} \mathbf{R}^{\frac{1}{2}} \tilde{\mathbf{V}} \tilde{\mathbf{V}}^\top. \end{aligned}$$

(iii) Express the basis projectors via factors B and A . Since B and A span the same spaces as U and V , we derive

$$\tilde{U}\tilde{U}^\top = B(B^\top L^{\frac{1}{2}}B)^{-1}B^\top, \quad \tilde{V}\tilde{V}^\top = A^\top(AR^{\frac{1}{2}}A^\top)^{-1}A.$$

Substituting these expressions into the formula for $\tilde{\mathcal{P}}_{\mathbb{T}_W}$, we obtain

$$\begin{aligned} \tilde{\mathcal{P}}_{\mathbb{T}_W}(Z) &= B(B^\top L^{\frac{1}{2}}B)^{-1}B^\top L^{\frac{1}{2}}Z + ZR^{\frac{1}{2}}A^\top(AR^{\frac{1}{2}}A^\top)^{-1}A \\ &\quad - B(B^\top L^{\frac{1}{2}}B)^{-1}B^\top L^{\frac{1}{2}}ZR^{\frac{1}{2}}A^\top(AR^{\frac{1}{2}}A^\top)^{-1}A. \end{aligned}$$

□

Proposition B.6. Suppose $W \in \mathcal{M}_r$ has a low-rank decomposition $W = BA$, where $B \in \mathbb{R}^{m \times r}$ and $A \in \mathbb{R}^{r \times n}$. For any matrix $M \in \mathbb{R}^{m \times r}$, $N \in \mathbb{R}^{r \times n}$, the matrix $MA + BN$ lies in the tangent space $\mathbb{T}_W$ at W of $\mathcal{M}_r$ at the point W .

Proof. Let $W \in \mathcal{M}_r$ has a compact singular value decomposition $W = U\Sigma V^\top$, where $U \in \mathbb{R}^{m \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$, and $V \in \mathbb{R}^{n \times r}$. By definition, the tangent space $\mathbb{T}_W$ at W is given by

$$\mathbb{T}_W = \{UK_1^\top + K_2V^\top \mid K_1 \in \mathbb{R}^{n \times r}, K_2 \in \mathbb{R}^{m \times r}\}.$$

Since B and A are low-rank factors of W , there exist invertible matrices $S \in \mathbb{R}^{r \times r}$ and $Q \in \mathbb{R}^{r \times r}$ such that

$$B = US, \quad A = QV^\top.$$

Substituting these expressions, the matrix $MA + BN$ can be rewritten as

$$MA + BN = MQV^\top + USN.$$

The first term, $MQV^\top$, lies in $\text{span}(V^\top)$, and the second term, USN , lies in $\text{span}(U)$. Thus, the sum $MQV^\top + USN$ lies in the tangent space $\mathbb{T}_W$ by the definition of the tangent space. Then, it follows that $MA + BN$ is on the tangent space $\mathbb{T}_W$. This completes the proof. □

B.5 Proof of Theorem 3.2 (closed-form factor update)

The proof has two stages: (i) solve (11) for $(\Delta_{B_t}, \Delta_{A_t})$ in terms of a free $X_t \in \mathbb{R}^{r \times r}$ (Lemma B.1); (ii) determine X_t by minimizing the $\mathcal{H}_t$ -imbalance from Solution 2 (Lemma B.2). Substituting (ii) into (i) yields the closed form in Theorem 3.2.

Lemma B.1 (X_t -parameterized factor solution). For problem (11), the optimal $(\Delta_{B_t}, \Delta_{A_t})$ are

$$\Delta_{B_t}^{\text{opt}} = (I - \tilde{P}_{B_t})L_t^{-1/2}G_{B_t}(A_tR_t^{1/2}A_t^\top)^{-1} - B_tX_t, \quad \Delta_{A_t}^{\text{opt}} = (B_t^\top L_t^{1/2}B_t)^{-1}G_{A_t}R_t^{-1/2} + X_tA_t,$$

where $X_t \in \mathbb{R}^{r \times r}$ is arbitrary.

Proof of Lemma B.1. Define

$$\Gamma(\Delta_{B_t}, \Delta_{A_t}) := \frac{1}{2} \|\Delta_{B_t}A_t + B_t\Delta_{A_t} - \tilde{\mathcal{P}}_{\mathbb{T}_t}(L_t^{-\frac{1}{2}}G_tR_t^{-\frac{1}{2}})\|_{\mathcal{H}_t}^2.$$

Differentiating $\Gamma(\Delta_{B_t}, \Delta_{A_t})$ with respect to Δ_{B_t} and Δ_{A_t} yields

$$\nabla_{\Delta_{B_t}} \Gamma(\Delta_{B_t}, \Delta_{A_t}) = L_t^{\frac{1}{2}}\Delta_{B_t}(A_tR_t^{\frac{1}{2}}A_t^\top) + L_t^{\frac{1}{2}}B_t\Delta_{A_t}R_t^{\frac{1}{2}}A_t^\top - G_tA_t^\top, \quad (20)$$

and

$$\nabla_{\Delta_{A_t}} \Gamma(\Delta_{B_t}, \Delta_{A_t}) = B_t^\top L_t^{\frac{1}{2}}\Delta_{B_t}A_tR_t^{\frac{1}{2}} + B_t^\top L_t^{\frac{1}{2}}B_t\Delta_{A_t}R_t^{\frac{1}{2}} - B_t^\top G_t. \quad (21)$$

Setting $\nabla_{\Delta_{B_t}} \Gamma(\Delta_{B_t}, \Delta_{A_t}) = 0$ and using the invertibility of $(A_tR_t^{\frac{1}{2}}A_t^\top)$ and $L_t^{\frac{1}{2}}$ gives

$$\Delta_{B_t} = L_t^{-\frac{1}{2}}G_tA_t^\top(A_tR_t^{\frac{1}{2}}A_t^\top)^{-1} - B_t\Delta_{A_t}R_t^{\frac{1}{2}}A_t^\top(A_tR_t^{\frac{1}{2}}A_t^\top)^{-1}. \quad (22)$$

Substituting (21) into $\nabla_{\Delta_{A_t}} \Gamma(\Delta_{B_t}, \Delta_{A_t}) = \mathbf{0}$ and using the invertibility of $B_t^\top L_t^{\frac{1}{2}} B_t$ and $R_t^{\frac{1}{2}}$ yields

$$\Delta_{A_t} [I - \tilde{Q}_{A_t}] = (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top G_t R_t^{-\frac{1}{2}} [I - \tilde{Q}_{A_t}],$$

where $\tilde{Q}_{A_t} = R_t^{\frac{1}{2}} A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} A_t$, which is the projection matrix onto the row space of A_t . Since $I - \tilde{Q}_{A_t}$ is the residual maker matrix, then a general solution is

$$\Delta_{A_t}^{\text{opt}} = (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top G_t R_t^{-\frac{1}{2}} + X_t A_t,$$

with arbitrary matrix $X_t \in \mathbb{R}^{r \times r}$. Plugging this Δ_{A_t} back into (22) gives

$$\Delta_{B_t}^{\text{opt}} = [I - \tilde{P}_{B_t}] L_t^{-\frac{1}{2}} G_t A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} - B_t X_t,$$

where $\tilde{P}_{B_t} = B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top L_t^{\frac{1}{2}}$, which is the projection matrix onto the column space of B_t . $\square$

This proves Lemma B.1. $\square$

Lemma B.2 ($\mathcal{H}_t$ -balance closed form). *The minimizer of $\frac{1}{2} \|\Delta_{B_t} A_t - B_t \Delta_{A_t}\|_{\mathcal{H}_t}^2$ over $X_t \in \mathbb{R}^{r \times r}$ (with $\Delta_{B_t}, \Delta_{A_t}$ given by Lemma B.1) is*

$$X_t^{\text{opt}} = -\frac{1}{2} (B_t^\top L_t^{1/2} B_t)^{-1} B_t^\top G_t A_t^\top (A_t R_t^{1/2} A_t^\top)^{-1}.$$

Substituting X_t^{opt} into Lemma B.1 yields the closed form in Theorem 3.2.

Proof of Lemma B.2. Let the objective function be $\Psi(X_t) = \frac{1}{2} \|\Delta_{B_t} A_t - B_t \Delta_{A_t}\|_{\mathcal{H}_t}^2$. To minimize $\Psi(X_t)$, we compute its gradient with respect to X_t ,

$$\nabla_{X_t} \Psi(X_t) = B_t^\top L_t^{\frac{1}{2}} (\Delta_{B_t} A_t - B_t \Delta_{A_t}) R_t^{\frac{1}{2}} A_t^\top.$$

Substituting the expressions for A_t and B_t from Theorem 3.2, we have

$$\begin{aligned} \nabla_{X_t} \Psi(X_t) &= B_t^\top L_t^{\frac{1}{2}} \left([I - B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top L_t^{\frac{1}{2}}] L_t^{-\frac{1}{2}} G_t A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} A_t \right. \\ &\quad \left. - B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top G_t R_t^{-\frac{1}{2}} - 2 B_t X_t A_t \right) R_t^{\frac{1}{2}} A_t^\top \\ &= -B_t^\top G_t A_t - 2 (B_t^\top L_t^{\frac{1}{2}} B_t) X_t (A_t R_t^{\frac{1}{2}} A_t^\top). \end{aligned}$$

Setting $\nabla_{X_t} \Psi(X_t) = \mathbf{0}$, we obtain

$$-B_t^\top G_t A_t = 2 (B_t^\top L_t^{\frac{1}{2}} B_t) X_t (A_t R_t^{\frac{1}{2}} A_t^\top).$$

Since $B_t^\top L_t^{\frac{1}{2}} B_t$ and $A_t R_t^{\frac{1}{2}} A_t^\top$ are invertible, we solve for X_t as

$$X_t^{\text{opt}} = -\frac{1}{2} (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top G_t A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1}.$$

This proves Lemma B.2. Substituting X_t^{opt} into Lemma B.1 and simplifying $-B_t X_t = \frac{1}{2} \tilde{P}_{B_t} L_t^{-1/2} G_t A_t^\top (A_t R_t^{1/2} A_t^\top)^{-1}$ (and the symmetric expression for $X_t A_t$) gives the closed form in Theorem 3.2, where $G_{B_t} = G_t A_t^\top$ and $G_{A_t} = B_t^\top G_t$ are the chain-rule low-rank gradients. $\square$

C Algorithm

In this section, we provide the completed algorithm of AdaPreLoRA.

Algorithm 1 AdaPreLoRA with SGD for fine-tuning.

- 1: Initialize $B_1 = \mathbf{0}_{m \times r}$, $A_1 = \text{Kaiming uniform}_{r \times n}$, $l_0 = \mathbf{0}_m$, $r_0 = \mathbf{0}_n$, $\epsilon = 1e - 6$.
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: $l_t = \beta_1 l_{t-1} + (1 - \beta_1) \sum_{j=1}^n (G_t \odot G_t)_{i,j}$, $L_t = \text{diag}(l_t / \sqrt{\|l_t\|_1})$.
 - 4: $r_t = \beta_2 r_{t-1} + (1 - \beta_2) \sum_{i=1}^m (G_t \odot G_t)_{i,j}$, $R_t = \text{diag}(r_t / \sqrt{\|r_t\|_1})$.
 - 5: $\Delta_{B_t} = \left[I - \frac{1}{2} B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top L_t^{\frac{1}{2}} \right] L_t^{-\frac{1}{2}} G_{B_t} (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1}$.
 - 6: $\Delta_{A_t} = (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} G_{A_t} R_t^{-\frac{1}{2}} \left[I - \frac{1}{2} R_t^{\frac{1}{2}} A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} A_t \right]$.
 - 7: $B_{t+1} = B_t - \eta_t \Delta_{B_t}$, $A_{t+1} = A_t - \eta_t \Delta_{A_t}$.
 - 8: **end for**
 - 9: Note: factor gradients $G_{B_t} = \partial \mathcal{L} / \partial B_t$ and $G_{A_t} = \partial \mathcal{L} / \partial A_t$ are obtained from autograd. The weight-space gradient G_t can be obtained either via a backward hook on W , or as the tangent-space surrogate $G_{B_t} A_t + B_t G_{A_t}$. Add ϵI to matrix $B_t^\top L_t^{\frac{1}{2}} B_t$ if it is not invertible.
-

Algorithm 2 AdaPreLoRA with Momentum for fine-tuning.

- 1: Initialize moment $M_0 = \mathbf{0}_{m \times n}$, $B_1 = \mathbf{0}_{m \times r}$, $A_1 = \text{Kaiming uniform}_{r \times n}$; $l_0 = \mathbf{0}_m$, $r_0 = \mathbf{0}_n$, weight decay λ , coefficients $\beta_1 = \beta_2$, and β_3 , $\epsilon = 1e - 6$.
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: $l_t = \beta_1 l_{t-1} + (1 - \beta_1) \sum_{j=1}^n (G_t \odot G_t)_{i,j}$, $L_t = \text{diag}(l_t / \sqrt{\|l_t\|_1})$.
 - 4: $r_t = \beta_2 r_{t-1} + (1 - \beta_2) \sum_{i=1}^m (G_t \odot G_t)_{i,j}$, $R_t = \text{diag}(r_t / \sqrt{\|r_t\|_1})$.
 - 5: $M_t = \beta_3 M_{t-1} + (1 - \beta_3) G_t$.
 - 6: $\Delta_{B_t} = \left[I - \frac{1}{2} B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top L_t^{\frac{1}{2}} \right] L_t^{-\frac{1}{2}} M_t A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1}$.
 - 7: $\Delta_{A_t} = (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top M_t R_t^{-\frac{1}{2}} \left[I - \frac{1}{2} R_t^{\frac{1}{2}} A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} A_t \right]$.
 - 8: $B_{t+1} = (1 - \lambda \eta_t) B_t - \eta_t \frac{\sqrt{1 - \beta_1^t}}{1 - \beta_3^t} \Delta_{B_t}$, $A_{t+1} = (1 - \lambda \eta_t) A_t - \eta_t \frac{\sqrt{1 - \beta_1^t}}{1 - \beta_3^t} \Delta_{A_t}$.
 - 9: **end for**
 - 10: Note: factor gradients $G_{B_t} = \partial \mathcal{L} / \partial B_t$ and $G_{A_t} = \partial \mathcal{L} / \partial A_t$ are obtained from autograd. The weight-space gradient G_t can be obtained either via a backward hook on W , or as the tangent-space surrogate $G_{B_t} A_t + B_t G_{A_t}$. Add ϵI to matrix $B_t^\top L_t^{\frac{1}{2}} B_t$ if it is not invertible. The momentum buffer M_t is maintained on G_t for theoretical optimality, but in practice we recommend maintaining first-order moments directly on the factor gradients G_{B_t} and G_{A_t} (then plugging the moment-debiased values into the $\Delta_{B_t} / \Delta_{A_t}$ formulas above), which avoids materializing an $m \times n$ moment buffer.
-

D Computational and Memory Complexity Analysis of SoLoRA

The update rule of SoLoRA is given by

$$\begin{aligned} \Delta_{A_t} &= (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} \underbrace{B_t^\top G_t}_{G_{A_t}} R_t^{-\frac{1}{2}} \left[I - \frac{1}{2} R_t^{\frac{1}{2}} A_t^\top (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1} A_t \right], \\ \Delta_{B_t} &= \left[I - \frac{1}{2} B_t (B_t^\top L_t^{\frac{1}{2}} B_t)^{-1} B_t^\top L_t^{\frac{1}{2}} \right] L_t^{-\frac{1}{2}} \underbrace{G_t A_t^\top}_{G_{B_t}} (A_t R_t^{\frac{1}{2}} A_t^\top)^{-1}. \end{aligned}$$

We now analyze the computational complexity of computing the updates Δ_{A_t} and Δ_{B_t} . For simplicity, we focus on Δ_{A_t} , as the complexity for Δ_{B_t} is symmetric.

- Compute gradient G_t . The stochastic gradient G_t of W_t is obtained during the backpropagation process.

- Row and column sums for $\mathbf{l}_t$ and $\mathbf{r}_t$. Compute $\mathbf{l}_t$ and $\mathbf{r}_t$ by summing the square of the element of $\mathbf{G}_t$ along rows or columns, which is in the computation $\mathcal{O}(mn)$. $\mathbf{L}_t^{\frac{1}{2}}$ and $\mathbf{L}_t^{-\frac{1}{2}}$ can be computed in $\mathcal{O}(m)$, $\mathbf{R}_t^{\frac{1}{2}}$ and $\mathbf{R}_t^{-\frac{1}{2}}$ can be computed in $\mathcal{O}(n)$.
- Compute $(\mathbf{B}_t^\top \mathbf{L}_t^{\frac{1}{2}} \mathbf{B}_t)^{-1} \mathbf{G}_{\mathbf{A}_t} \mathbf{R}_t^{-\frac{1}{2}}$. First to compute the inverse matrices $(\mathbf{B}_t^\top \mathbf{L}_t^{\frac{1}{2}} \mathbf{B}_t)^{-1}$ in $\mathcal{O}((m+r)r^2)$. Then multiply the inverse $(\mathbf{B}_t^\top \mathbf{L}_t^{\frac{1}{2}} \mathbf{B}_t)^{-1}$ by $\mathbf{G}_{\mathbf{A}_t}$ in $\mathcal{O}(nr^2)$, and multiply the diagonal matrix $\mathbf{R}_t^{-\frac{1}{2}}$ in $\mathcal{O}(nr)$.
- Compute $(\mathbf{B}_t^\top \mathbf{L}_t^{\frac{1}{2}} \mathbf{B}_t)^{-1} \mathbf{G}_{\mathbf{A}_t} \mathbf{A}_t^\top (\mathbf{A}_t \mathbf{R}_t^{\frac{1}{2}} \mathbf{A}_t^\top)^{-1} \mathbf{A}_t$. First to compute the inverse matrices $(\mathbf{A}_t \mathbf{R}_t^{\frac{1}{2}} \mathbf{A}_t^\top)^{-1}$ in $\mathcal{O}((n+r)r^2)$. Use the result from the last step, multiply $(\mathbf{B}_t^\top \mathbf{L}_t^{\frac{1}{2}} \mathbf{B}_t)^{-1} \mathbf{G}_{\mathbf{A}_t}$ by $\mathbf{A}_t^\top$ in computation $\mathcal{O}(nr^2)$, then multiply $(\mathbf{A}_t \mathbf{R}_t^{\frac{1}{2}} \mathbf{A}_t^\top)^{-1}$ in computation $\mathcal{O}(r^3)$, and multiply $\mathbf{A}_t$ in $\mathcal{O}(nr^2)$.

The computation complexity of $\Delta_{\mathbf{A}_t}$ is $\mathcal{O}(mn + (m+n)r^2 + r^3)$. The computation of $\Delta_{\mathbf{B}_t}$ follows a similar structure, with symmetric terms. Its complexity is also $\mathcal{O}(mn + (m+n)r^2 + r^3)$. Then we have

- **Per Iteration Computational Complexity.** Combining the computations of $\Delta_{\mathbf{A}_t}$ and $\Delta_{\mathbf{B}_t}$, the total computation complexity per iteration is $\mathcal{O}(mn + (m+n)r^2 + r^3)$.
- **Memory Complexity.** The algorithm requires storing the vectors $\mathbf{l}_t$ and $\mathbf{r}_t$ in each iteration, hence the memory complexity is $\mathcal{O}(m+n)$.

E Supplementary Experiments of GPT-2 Fine-tuning

E.1 Parameter Settings

To ensure the reproducibility of the experiments described in Section 4 and to facilitate verification and comparison by others, we provide the complete details of the experimental parameter settings. Tables 8 and 9 list the parameters used during the fine-tuning of GPT-2 models and the learning rates corresponding to different optimizers, respectively. Specifically, we conduct experiments with GPT-2 models of various sizes. ‘‘Rank 4 (M)’’ represents a medium-sized model using LoRA with rank 4, while ‘‘Rank 4’’, ‘‘Rank 16’’, and ‘‘Rank 64’’ represent small models using LoRA with ranks 4, 16, and 64, respectively. To ensure the fairness of the experimental setup, we follow the parameter settings in LoRA [14] and Riemannian Preconditioned LoRA [37]. However, considering the sensitivity of different optimizers to learning rates, we use a grid search strategy to independently tune the optimal learning rate for each optimizer. This ensures that each optimizer operates under its best-performing configuration, providing more objective and reliable experimental results.

Table 8: Training and Inference Configuration for GPT-2 Fine-tuning.

Training		LoRA α		Inference	
Parameter	Value	Parameter	Value	Parameter	Value
Dropout Probability	0.1				
Batch Size	8				
Number of Epochs	5	α (for Rank 4)	32	Beam Size	10
Warm-up Steps	500	α (for Rank 16)	32	Length Penalty	0.8
Learning Rate Scheduler	Linear	α (for Rank 64)	128	No Repeat Ngram Size	4
Label Smoothing	0.1				
Weight Decay	0.01				

E.2 Cross-rank ablation on GPT-2 small

To complement the $r = 4$ comparisons in the main text (Table 2), we report cross-rank scores at $r \in \{16, 64\}$ on GPT-2 small in Table 10. Hyperparameters follow Table 8 and Table 9. Across both

Table 9: Core Optimizer Parameters for GPT-2 fine-tuning.

Methods	Learning Rate ($\times 10^{-3}$)				β_3	$\beta_1 = \beta_2$
	Rank 4	Rank 4 (M)	Rank 16	Rank 64		
SGD	90	90	200	90	/	/
Scaled GD	20	20	40	10	/	/
LoRA-Pro SGD	40	40	40	40	/	/
AdaPreLoRA SGD	0.05	0.05	0.5	0.8	/	0.98
AdamW	0.2	0.2	0.2	0.2	0.9	0.999
Scaled AdamW	0.8	0.8	2	4	0.7	0.8
LoRA-Pro AdamW	0.1	0.1	0.2	0.4	0.9	0.999
AdaPreLoRA AdamW	0.5	0.1	0.8	0.3	0.9	0.98

ranks and both optimizer families, AdaPreLoRA achieves the best or tied-best score on essentially every metric; gains over Scaled GD/AdamW and LoRA-Pro narrow at higher rank but remain positive, consistent with the main-text observation that the gradient-statistics-aware projection retains an advantage even when the LoRA factorization is less rank-constrained.

Table 10: GPT-2 small fine-tuned on E2E at $r \in \{16, 64\}$ (cross-rank ablation). Bold/underline = best/second-best per metric per (rank, optimizer family).

r	Method	E2E				
		BLEU	NIST	MET	ROUGE-L	CIDEr
16	SGD	65.4	8.07	40.7	67.0	2.07
	Scaled GD	<u>68.8</u>	<u>8.75</u>	45.0	69.2	<u>2.39</u>
	LoRA-Pro SGD	68.3	8.67	<u>45.1</u>	<u>69.3</u>	2.37
	AdaPreLoRA SGD (ours)	70.0	8.82	46.6	71.6	2.53
	AdamW	69.5	8.77	46.4	71.2	2.48
	Scaled AdamW	<u>69.8</u>	<u>8.79</u>	46.5	<u>71.7</u>	<u>2.51</u>
	LoRA-Pro AdamW	69.7	8.73	46.8	<u>71.7</u>	<u>2.51</u>
	AdaPreLoRA AdamW (ours)	70.2	8.85	<u>46.6</u>	71.9	2.52
64	SGD	64.7	8.08	40.8	66.7	2.04
	Scaled GD	68.5	8.68	45.0	69.4	<u>2.38</u>
	LoRA-Pro SGD	<u>68.6</u>	<u>8.71</u>	<u>45.4</u>	<u>69.7</u>	<u>2.38</u>
	AdaPreLoRA SGD (ours)	70.1	8.85	46.7	71.8	2.53
	AdamW	69.6	8.76	<u>46.7</u>	<u>71.5</u>	2.50
	Scaled AdamW	<u>70.0</u>	<u>8.83</u>	46.4	<u>71.5</u>	2.50
	LoRA-Pro AdamW	<u>70.0</u>	8.82	46.6	<u>71.5</u>	<u>2.51</u>
	AdaPreLoRA AdamW (ours)	70.2	8.84	46.8	72.1	2.52

E.3 Training Efficiency Comparison

To validate the training and inference efficiency of AdaPreLoRA, we report in Table 11 the total training and inference time required for all algorithms on the GPT-2 small model (rank 64) fine-tuned on E2E.

F Supplementary Experiments of Diffusion Model Fine-tuning

We evaluate AdaPreLoRA on diffusion-model personalization with the Mix-of-Show framework [9], which integrates Embedding Decomposed LoRA (EDLoRA) into the text encoder and U-Net of a Stable Diffusion backbone. Following [9, 37], we disable embedding-vector tuning and fine-tune only the LoRA components of the text encoder and U-Net submodules. The CLIP score [12] (ViT-B/32 variant of the CLIP model [27]) measures alignment between generated images and text prompts (range [0, 100], higher better); FID [13] measures the distributional similarity between generated and

Table 11: Training and inference time (hours) of GPT-2 small (rank=64) fine-tuned with different optimizers on E2E.

Method	SGD	Scaled GD	LoRA-Pro SGD	AdaPreLoRA SGD
Training	1.79	1.92	2.78	2.04
Inference	1.86	1.87	1.58	1.89
Method	AdamW	Scaled AdamW	LoRA-Pro AdamW	AdaPreLoRA AdamW
Training	1.79	1.94	2.93	2.04
Inference	1.87	1.88	1.89	1.86

Table 12: Optimizer Parameters for fine-tuning the Mix-of-Show Model.

Methods	Learning Rate		β_3	$\beta_1 = \beta_2$
	Text-Encoder	U-Net		
SGD	1e-1	1e-1	/	/
Scaled GD	1e-1	1e-1	/	/
LoRA-Pro SGD	1e-1	1e-1	/	/
AdaPreLoRA SGD	1e-5	1e-5	/	0.98
AdamW	1e-5	1e-4	0.9	0.999
Scaled AdamW	1e-5	1e-4	0.7	0.8
LoRA-Pro AdamW	1e-5	1e-5	0.9	0.999
AdaPreLoRA AdamW	1e-5	1e-5	0.9	0.98

reference images (lower better). Aggregate CLIP/FID scores at LoRA scaling factors $\{0.7, 1.0\}$ are in the main text (Table 6); this section reports per-prompt qualitative samples at the same two scaling factors, using the per-optimizer learning rates listed in Table 12. At $s = 1.0$, Figures 2 and 3 compare AdamW-based optimizers on Harry Potter and Hermione Granger prompts respectively, and Figures 4 and 5 compare SGD-based optimizers on the same prompts. At $s = 0.7$, Figures 6 and 7 report the corresponding AdamW comparisons. Across both scaling factors, AdaPreLoRA preserves character identity and follows the prompt scene layout while remaining visually consistent across prompts.

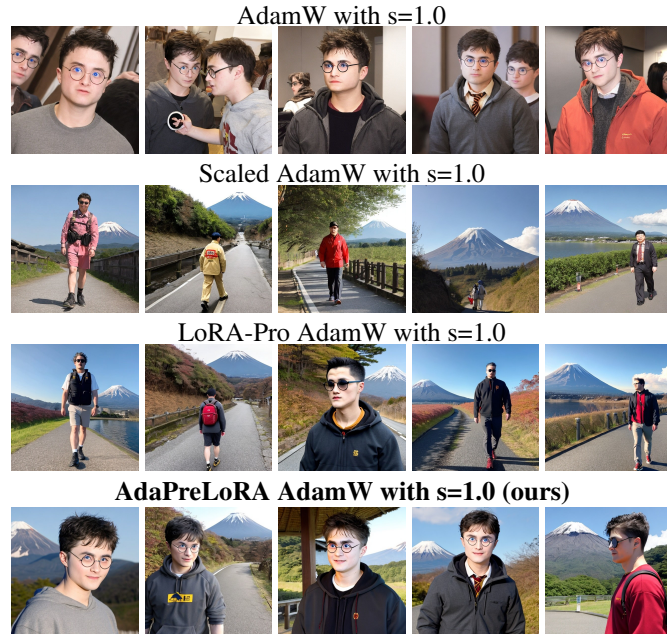


Figure 2: Generated results based on the prompt “Harry Potter is walking near Mount Fuji” when fine-tuned using AdamW-based optimizers. All optimizers employed a LoRA scaling factor of 1.0, with the best learning rate. The results indicate that the output of the model trained with our optimizer incorporates the character “Harry Potter”, the action “walking”, and the scene “Mount Fuji”, yielding superior image quality compared to alternative approaches.



Figure 3: Generation results from the prompt “A photo of Hermione Granger on the beach, small waves, detailed symmetric face, beautiful composition” using AdamW-based optimizers. All the optimizers apply LoRA scaling factor as 1.0, with the best learning rate. Results demonstrate that the model trained with our optimizer generates higher-quality images than others, especially the face of Hermione Granger and the scene.



Figure 4: Generated results based on the prompt “Harry Potter standing near the lake” when fine-tuned using SGD-based optimizers. All optimizers employed a LoRA scaling factor of 1.0, with the best learning rate. Results demonstrate that the output images of the model trained with our optimizer have higher-quality than others, especially the face of Harry Potter.



Figure 5: Generated results based on the prompt “Hermione Granger wearing a brown shirt” when fine-tuned using SGD-based optimizers. All optimizers employed a LoRA scaling factor of 1.0, with the best learning rate. Results demonstrate that the model trained with AdaPreLoRA generates higher-quality images than others, especially the face of Hermione Granger.

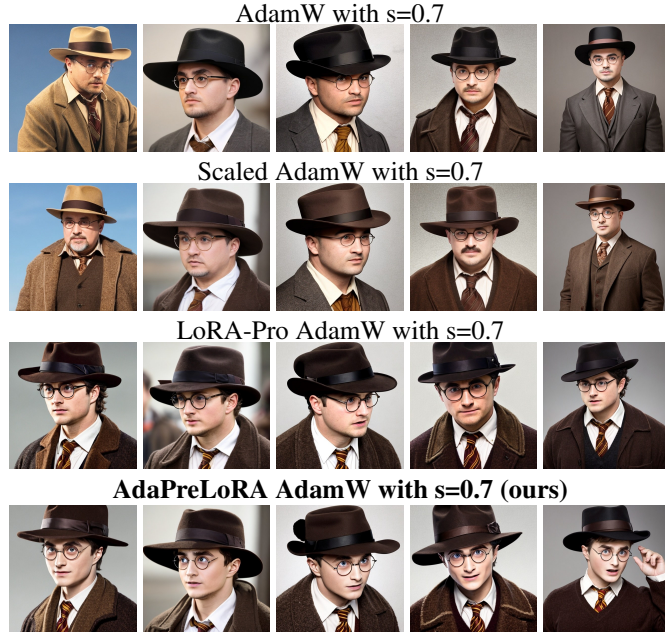


Figure 6: Generated results based on the prompt “Harry Potter wearing a brown hat” when fine-tuned using AdamW-based optimizers. All optimizers employed a LoRA scaling factor of 0.7, with the best learning rate. The results indicate that the output of the model trained with AdaPreLoRA incorporates the character “Harry Potter”, and the “hat”, yielding superior quality compared to alternative approaches.



Figure 7: Generation results from the prompt “A photo of Hermione Granger on the beach, small waves, detailed symmetric face, beautiful composition” using AdamW-based optimizers. All the optimizers apply LoRA scaling factor as 0.7. Results demonstrate that AdaPreLoRA generates higher-quality images for both scaling factors than others, including the face of Hermione Granger and the scene.